PRUNING FREE RESOLUTIONS OVER A COMPLETE INTERSECTION

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ABSTRACT. Given a free resolution $F_\bullet$ over a polynomial ring R and a system of higher homotopies on it for a regular sequence $a_1, \dots, a_c$, we can use the Eisenbud-Shamash construction to build a free resolution over $\bar{R} = R/(a_1, \dots, a_c)$. We provide a combinatorial formula for the new system of higher homotopies on a pruned free resolution $F_\bullet{}^{\mathcal{M}}$ that corresponds to a pruning of the Eisenbud-Shamash resolution.

INTRODUCTION

Free resolutions over a complete intersection are typically infinite, and the standard way to build one, due to Shamash [26] and Eisenbud [16], starts from a finite free resolution over a polynomial ring. One such free resolution is the Taylor resolution [30] of a monomial ideal over a polynomial ring. In [29], Sobieska gives a compact formula for the Taylor resolution over a complete intersection using the construction of Eisenbud and Shamash. Unfortunately, Taylor resolutions are rarely minimal. Over the polynomial ring, algebraic Morse theory is a standard tool for pruning a resolution down toward the minimal one, although, sometimes there are obstructions to achieving minimality. There are two main algorithms which produce Morse matchings on the Taylor resolution of a monomial ideal: the pruning algorithm [1] and the Barile–Macchia algorithm [3, 11]. Several authors have exploited these algorithms to produce minimal resolutions for monomial ideals with five or fewer generators [18, 8], which are edge ideals of certain graphs [3, 11, 12, 10], powers of monomial ideals which have minimal pruned resolutions [13, 14, 9] or ideals where the Morse matching produces nice DG structure [6, 20]. We show that algebraic Morse theory can also be used to prune these infinite resolutions over complete intersections.

Let R be a polynomial ring over a field, $F_\bullet$ a free resolution of R/I over R with differential d, and $a_1, \dots, a_c$ a regular sequence contained in I, with $\overline{R} = R/(a_1, \dots, a_c)$. A *system of higher homotopies* σ is a family of maps $\sigma_{\boldsymbol{u}}$ on $F_\bullet$, for $\boldsymbol{u} \in \mathbb{N}^c$, raising homological degree by $2|\boldsymbol{u}| - 1$, with $\sigma_{\mathbf{0}} = d_\bullet$ and $\sum_{\boldsymbol{v}+\boldsymbol{v}'=\boldsymbol{u}} \sigma_{\boldsymbol{v}}\sigma_{\boldsymbol{v}'}$ equal to $a_j \operatorname{id}$ if $\boldsymbol{u} = \boldsymbol{e}_j$ and 0 otherwise. It is *truncated* if $\sigma_{\boldsymbol{u}} = 0$ for $|\boldsymbol{u}| \geq 2$. The *Eisenbud-Shamash resolution* $\operatorname{Sh}(F_\bullet, \sigma)$ is free over $\overline{R}$ on the cells $y^{\boldsymbol{u}} \otimes \alpha \otimes \overline{1}$, for $\boldsymbol{u} \in \mathbb{N}^c$ and α in the basis of $F_\bullet$, with differential $\sum_{\boldsymbol{u}} t^{\boldsymbol{u}} \otimes \sigma_{\boldsymbol{u}} \otimes \operatorname{id}$, where the *CI operator* t_j sends $y^{\boldsymbol{u}}$ to $y^{\boldsymbol{u}-\boldsymbol{e}_j}$.

Algebraic Morse theory is due independently to Sköldberg [27] and to Jöllenbeck and Welker [23], building on Forman's discrete Morse theory [19] and its use on cellular resolutions by Batzies and Welker [4] and by Chari [7]. The *complex graph* $G(F_\bullet)$ has the basis elements of $F_\bullet$, the *cells*, as vertices and an edge $\alpha \to \beta$ whenever the coefficient $d_{\alpha,\beta}$ of β in $d(\alpha)$ is nonzero. A *Morse matching* $\mathcal{M}$ is a set of edges with unit coefficient, no two sharing a cell, such that reversing them leaves no infinite directed path. The *Morse complex* $F_\bullet{}^{\mathcal{M}}$ is free on the unmatched, *critical*, cells, with differential given by sums over paths, and a contraction $(i^{\mathcal{M}}, p^{\mathcal{M}}, h^{\mathcal{M}})$ which makes it a deformation retract of $F_\bullet$ (Theorem 6.8).

For a truncated system $\sigma = \{\sigma^{(j)}\}_{j=1}^c$, with $\sigma^{(j)} = \sigma_{\boldsymbol{e}_j}$, and a Morse matching $\mathcal{M}$ on $F_\bullet$, let $\mathscr{M}$ be one copy of $\mathcal{M}$ over each $\boldsymbol{u} \in \mathbb{N}^c$. The main result (Theorem 9.1) is that, under the identity

on the common basis,

$$\mathrm{Sh}(F_\bullet, \sigma)^{\mathcal{M}} = \mathrm{Sh}\big(F_\bullet{}^{\mathcal{M}}, \sigma^{\mathcal{M}}\big), \qquad \sigma^{\mathcal{M}}_{\boldsymbol{u}} = \sum_{\boldsymbol{e}_{j_1}+\cdots+\boldsymbol{e}_{j_k}=\boldsymbol{u}} p^{\mathcal{M}}\sigma^{(j_1)}h^{\mathcal{M}}\sigma^{(j_2)}\cdots h^{\mathcal{M}}\sigma^{(j_k)}i^{\mathcal{M}},$$

and that $\sigma^{\mathcal{M}}$ is again a system of higher homotopies, in general no longer truncated.

The proof applies the homological perturbation lemma, which goes back to Brown [5] and Gugenheim [21] (see Crainic [15] for a modern account). Sköldberg [28] derives algebraic Morse theory from it for a filtered complex. We replace his filtration by the assumption that the relevant geometric series is finite on each element (Lemma 6.7), and we apply the lemma twice and use an additivity property of perturbations (Lemma 9.7). We split the Shamash differential into the part along each copy of $F_\bullet$ and the connecting part ∂_{con} between copies, and perturb first by the relevant perturbation at each level and then by the connecting part. Lemma 9.7, the Perturbation Additivity Lemma, is the main tool to show that performing two perturbations produces the same resolution as doing the single perturbation by the sum of the perturbations. Lemma 9.7 is a version of a result of Real [25, Proposition 3.6] under weaker hypotheses on the second perturbation, and our main theorem puts it to significant use.

The paper is structured as follows. We fix notation in Section 1 for easy reference. In Section 2, we recall the Eisenbud-Shamash resolution. In Sections 3 and 4, we continue with some insights on how to decompose the Shamash differentials and view them as block matrices. In Section 5 we start our running example with the Taylor resolution in Example 5.3 and recall Sobieska's homotopies for the Taylor resolution of a complete intersection. In Section 6, we introduce algebraic Morse Theory and continue our running example with matchings over a polynomial ring in Example 6.9. In Section 7, we define lifted matchings to the Eisenbud-Shamash resolution for any matching on the Taylor resolution; we prove that these lifted matchings are Morse when the original matching is. In Section 8 as preparation for our Main Theorem 9.1, we present some motivating examples. The paper concludes with Section 9, where we prove the main theorem, Theorem 9.1. We also produce an example where the higher homotopies of the lifted matching are no longer truncated even though those of the Taylor resolution were.

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1. Notation

Throughout, A will be a commutative Noetherian ring, $R = \Bbbk[x_1, \ldots, x_n]$ will be a polynomial ring over a field $\Bbbk$ and $\overline{R} = R/\mathfrak{a}$ where $\mathfrak{a} = (a_1, \ldots, a_c)$ and $a_1, \ldots, a_c$ forms a regular sequence in R.

This paper pertains to complexes of finite free A-modules: $(F_\bullet, d_\bullet)$ where F_q is free of rank s_q for some $s_q \in \mathbb{N}$ with basis $\mathcal{B}_q$ and $d_q : F_q \to F_{q-1}$ can be represented by an $s_{q-1} \times s_q$ matrix with an appropriate choice of basis. We call elements $x \in F_q$, elements of *homological degree* q and we will write $\operatorname{hdeg} x = q$. The basis for the complex is $\mathcal{B} = \bigcup_q \mathcal{B}_q$ so that d has matrix representation from $\mathcal{B}$.

For free complexes $(F_\bullet, d)$ and $(F'_\bullet, d')$, we will consider various collections of maps between them, which sometimes have adornments such as subscripts or superscripts. Because these adornments are important descriptors of these maps, we will use the convention that if $f : F_\bullet \to F'_\bullet$ is such a collection of maps, possibly not a chain map, possibly of nonzero external degree, $f|_q$ will refer to the map $f_\bullet$ which starts at the homological degree q in $F_\bullet$. Sometimes, however, it will be clear from the representation of our elements in F_q in terms of its basis elements which homological degree the map starts in; in this case we will drop $|_q$.

We use the standard fine multigrading for elements in a polynomial ring, $R = \Bbbk[x_1, \dots, x_n]$. The *homogeneous elements* are the terms $rx_1^{b_1} \cdots x_n^{b_n}$, where $r \in \Bbbk$ and $(b_1, \dots, b_n) \in \mathbb{N}^n$. If r is a homogeneous element of R of multidegree $\boldsymbol{b} = (b_1, \dots, b_n)$, we will write $\operatorname{gr}(r) = \boldsymbol{b}$. $\mathbb{N}^n$ is endowed with the componentwise partial order, so $\boldsymbol{b} \le \boldsymbol{b'}$ means $b_i \le b'_i$ for every $1 \le i \le n$, and $\boldsymbol{b} < \boldsymbol{b'}$ means $\boldsymbol{b} \le \boldsymbol{b'}$ with $\boldsymbol{b} \ne \boldsymbol{b'}$. For $\boldsymbol{u} \in \mathbb{N}^n$ we write $|\boldsymbol{u}| = u_1 + \cdots + u_n$, and $\boldsymbol{e}_1, \dots, \boldsymbol{e}_n$ are the standard basis vectors of $\mathbb{N}^n$. Sometimes we will represent the multidegree of an element by the monomial $x^{\boldsymbol{b}} = x_1^{b_1} \cdots x_n^{b_n}$ instead of by $\boldsymbol{b}$. Then $\boldsymbol{b} \le \boldsymbol{b'}$ if and only if $x^{\boldsymbol{b}} \mid x^{\boldsymbol{b'}}$.

We have compiled a list of notation we use in the paper and where we first reference.

List of notation.

- $(F_\bullet, d)$: free complex with basis $\mathcal{B} = \bigcup_n \mathcal{B}_n$ where $\mathcal{B}_n$ is the basis of F_n. (Section 1, Definition 6.1)
- $f|_q$: f restricted to F_q. (Section 1)
- $|\boldsymbol{u}| = u_1 + \cdots + u_c$ where $\boldsymbol{u} = (u_1, \dots, u_c)$. (Section 1)
- $\boldsymbol{e}_j$ the jth standard basis vector. (Section 1)
- t_j: CI operator of degree -2, $t_j(y^{\boldsymbol{v}}) = y^{\boldsymbol{v}-\boldsymbol{e}_j}$; $t^{\boldsymbol{u}} = t_1^{u_1} \cdots t_c^{u_c}$. (Definition 2.1)
- $R\{y_j\}_{j=1}^c = \bigoplus_{\boldsymbol{u}} Ry^{\boldsymbol{u}}$, $y^{\boldsymbol{u}} = y_1^{(u_1)} \cdots y_c^{(u_c)}$ of degree $2|\boldsymbol{u}|$. (Definition 2.1)
- $\sigma = \{\sigma_{\boldsymbol{u}}\}$: system of higher homotopies, $\sigma_{\boldsymbol{0}} = d$; $\sigma^{(j)} = \sigma_{\boldsymbol{e}_j}$. (Definition 2.2)
- $\operatorname{Sh}(F_\bullet, \sigma) = R\{y_j\}_{j=1}^c \otimes_R F_\bullet \otimes_R \overline{R}$, $d_{\operatorname{Sh}} = \sum_{\boldsymbol{u}} t^{\boldsymbol{u}} \otimes \sigma_{\boldsymbol{u}} \otimes \mathrm{id} = \overline{D}$, where $D = \sum_{\boldsymbol{u}} t^{\boldsymbol{u}} \otimes \sigma_{\boldsymbol{u}}$ is a map of $R\{y_j\}_{j=1}^c \otimes_R F_\bullet$. (Definition 2.4)
- $\overline{F}_\bullet = F_\bullet \otimes_R \overline{R}$ with basis $\bar{\alpha} = \alpha \otimes \overline{1}$, and $\bar{g} = g \otimes \mathrm{id}$ for an R-linear map g: a bar is reduction modulo $\mathfrak{a}$. (Definition 2.4)
- $y^{\boldsymbol{u}}\alpha$: the cell $y^{\boldsymbol{u}} \otimes \bar{\alpha}$ of $\operatorname{Sh}(F_\bullet, \sigma)$. In matrices, examples and figures we omit the tensor sign and the bar, write α for $1 \otimes \bar{\alpha}$, and write $y^{\boldsymbol{u}}F_\bullet$ for the copy $y^{\boldsymbol{u}} \otimes \overline{F}_\bullet$. (Section 3, Section 7)
- $\partial_0 = \mathrm{id} \otimes d$, $\partial_j = t_j \otimes \sigma^{(j)}$, $\partial_{\mathrm{con}} = \partial_1 + \cdots + \partial_c$, maps of $R\{y_j\}_{j=1}^c \otimes_R F_\bullet$, with $D = \partial_0 + \partial_{\mathrm{con}}$ and $d_{\operatorname{Sh}} = \bar{\partial}_0 + \bar{\partial}_{\mathrm{con}}$. (Remark 3.2)
- $\Theta = \sum_j t_j \otimes a_j \,\mathrm{id}$, equal to D^2. (Remark 3.2)
- $N_k = \binom{k+c-1}{c-1}$, the number of $\boldsymbol{u}$ with $|\boldsymbol{u}| = k$. (Section 4)
- $T_{\boldsymbol{v}}^{(k)}$: matrix of $t^{\boldsymbol{v}}$ from level k to level $k - |\boldsymbol{v}|$; $S_k = \sum_j T_{\boldsymbol{e}_j}^{(k)} \otimes \sigma^{(j)}$. (Section 4)
- ε_S: Taylor cell, $S \subseteq \{1, \dots, r\}$, $\operatorname{hdeg} \varepsilon_S = |S|$, $m_S = \operatorname{lcm}(m_i : i \in S) = \operatorname{gr}(\varepsilon_S)$. (Section 5)
- $d_{\alpha,\beta}$: the coefficient of β in $d_q(\alpha)$ where $\alpha \in \mathcal{B}_q$ and $\beta \in \mathcal{B}_{q-1}$, the (β, α) entry of matrix d_q. (Definition 6.1)
- $f_{\alpha,\beta}$: the coefficient of β in $f|_q(\alpha)$ where $\alpha, \beta \in \mathcal{B}$. (Definition 6.1)
- $G(F_\bullet)$: weighted directed graph vertices $\mathcal{B}$, edges $\alpha \to \beta$ of weight $d_{\alpha,\beta} \ne 0$. (Definition 6.3)
- $\mathcal{M}$: partial matching of unit edges of $G(F_\bullet)$. (Definition 6.3)
- $G^{\mathrm{rep}}(F_\bullet, \mathcal{M})$: $G(F_\bullet)$ with each $(\alpha, \beta) \in \mathcal{M}$ reversed, of weight $-(d_{\alpha,\beta})^{-1}$. (Definition 6.3)
- μ: the entries $d_{\alpha,\beta}$ with $(\alpha, \beta) \in \mathcal{M}$; $\delta = d - \mu$. (Section 6, just before Definition 6.6)
- (i, p, h): contraction, $pi = \mathrm{id}$, $ip - \mathrm{id} = dh + hd$, $h^2 = hi = ph = 0$. (Definition 6.6)
- $\Phi = \sum_{n \ge 0} (h\delta)^n = (\mathrm{id} - h\delta)^{-1}$. (Section 6, just after Definition 6.6)
- $\Gamma(\alpha, \beta) = \sum_\pi \operatorname{wt}(\pi)$ over gradient paths π from α to β. (Section 6, just before Theorem 6.8)
- $d^{\mathcal{M}} = p\delta\Phi i$, $d^{\mathcal{M}}_{\alpha,\beta} = \Gamma(\alpha, \beta)$; $F_\bullet{}^{\mathcal{M}} = (K_\bullet, d^{\mathcal{M}})$. (Theorem 6.8)
- $(i^{\mathcal{M}}, p^{\mathcal{M}}, h^{\mathcal{M}}) = (\Phi i,\ p + p\delta\Phi h,\ \Phi h)$. (Theorem 6.8)
- $\mathcal{M} = \{(y^{\boldsymbol{u}} \otimes \bar{\alpha}, y^{\boldsymbol{u}} \otimes \bar{\beta}) : (\alpha, \beta) \in \mathcal{M}\}$. (Definition 7.2)
- $\sigma_{\boldsymbol{0}}^{\mathcal{M}} = d^{\mathcal{M}}$, $\sigma_{\boldsymbol{u}}^{\mathcal{M}} = \sum p^{\mathcal{M}} \sigma^{(j_1)} h^{\mathcal{M}} \cdots h^{\mathcal{M}} \sigma^{(j_k)} i^{\mathcal{M}}$ over $\boldsymbol{e}_{j_1} + \cdots + \boldsymbol{e}_{j_k} = \boldsymbol{u}$. (Definition 7.5)
- $f = \mathrm{id} \otimes f$, a map of $R\{y_j\}_{j=1}^c \otimes_R F_\bullet$, so $f(y^{\boldsymbol{u}} \otimes \alpha) = y^{\boldsymbol{u}} \otimes f(\alpha)$. On $\operatorname{Sh}(F_\bullet, \sigma)$ we use $\overline{f}$. For the Greek maps δ and μ we write $\mathrm{id} \otimes \delta$ and $\mathrm{id} \otimes \mu$. (Section 9, just before Proposition 9.4)

2. The Eisenbud-Shamash construction

This section recalls the Shamash construction. The construction, with the divided powers below, is Shamash's [26] for a single element a_1. Systems of higher homotopies for a regular sequence and the operators t_j were introduced by Eisenbud in [16], the operators having appeared earlier on Ext in Gulliksen's work [22] and are described combinatorially in Section 4.

Definition 2.1. [16] Let $R[t_j]_{j=1}^c = R[t_1, \dots, t_c]$ be the polynomial ring over R with additional variables t_j in degree -2. Let $R\{y_j\}_{j=1}^c = \bigoplus_{\boldsymbol{u}\in\mathbb{N}^c} Ry^{\boldsymbol{u}}$ be its graded dual, the free R-module on the symbols $y^{\boldsymbol{u}} = y_1^{(u_1)} \cdots y_c^{(u_c)}$ dual to the monomials $t^{\boldsymbol{u}}$, for $\boldsymbol{u} \in \mathbb{N}^c$, with $y^{\boldsymbol{u}}$ in homological degree $2|\boldsymbol{u}|$.

The dual of an $R[t_j]_{j=1}^c$-module is an $R[t_j]_{j=1}^c$-module by $(t_j\varphi)(g) = \varphi(t_j g)$, for φ in the dual and g in the module. In the dual basis $t_j(y^{\boldsymbol{v}}) = y^{\boldsymbol{v}-\boldsymbol{e}_j}$ when $v_j \geq 1$ and $t_j(y^{\boldsymbol{v}}) = 0$ when $v_j = 0$. Following [17] we call t_j, acting this way, the *CI operator* t_j.

The CI operators commute. For $\boldsymbol{u} \in \mathbb{N}^c$ the composite $t^{\boldsymbol{u}} = t_1^{u_1} \cdots t_c^{u_c}$ satisfies $t^{\boldsymbol{u}}(y^{\boldsymbol{v}}) = y^{\boldsymbol{v}-\boldsymbol{u}}$ when $\boldsymbol{u} \leq \boldsymbol{v}$ and $t^{\boldsymbol{u}}(y^{\boldsymbol{v}}) = 0$ otherwise. Then $t^{\boldsymbol{v}}t^{\boldsymbol{v'}} = t^{\boldsymbol{v}+\boldsymbol{v'}}$ for all $\boldsymbol{v}, \boldsymbol{v'} \in \mathbb{N}^c$.

We call an index $j \in \{1, \dots, c\}$ a *color*. We will later represent our base free resolution by a weighted directed graph. We will have c homotopies, each corresponding to added arrows on the graph, and so we color the arrows to distinguish which homotopy they describe.

The module $R\{y_j\}_{j=1}^c$ is the divided power algebra on $y_1, \dots, y_c$, but only its module structure and the $R[t_j]_{j=1}^c$-action above are used here. The CI operators are also called Eisenbud's operators or cohomological operators [2]. Every $y^{\boldsymbol{u}}$ has even homological degree, so tensor factors from $R\{y_j\}_{j=1}^c$ contribute no Koszul signs anywhere below.

Definition 2.2. Let $F_\bullet$ be a free resolution over R and $a_1, \dots, a_c \in R$. A *system of higher homotopies* for $a_1, \dots, a_c$ on $F_\bullet$ is a family $\{\sigma_{\boldsymbol{u}}\}_{\boldsymbol{u}\in\mathbb{N}^c}$ of R-linear maps $\sigma_{\boldsymbol{u}} \colon F_\bullet \to F_{\bullet+2|\boldsymbol{u}|-1}$ with $\sigma_{\boldsymbol{0}} = d$ and

$$\sum_{\boldsymbol{v}+\boldsymbol{v'}=\boldsymbol{u}} \sigma_{\boldsymbol{v}}\sigma_{\boldsymbol{v'}} = \begin{cases} a_j \operatorname{id} & \text{if } \boldsymbol{u} = \boldsymbol{e}_j, \\ 0 & \text{otherwise.} \end{cases} \tag{1}$$

A system of higher homotopies is *truncated* if $\sigma_{\boldsymbol{u}} = 0$ for all $\boldsymbol{u}$ with $|\boldsymbol{u}| \geq 2$. Write $\sigma^{(j)} = \sigma_{\boldsymbol{e}_j}$. For a truncated system, (1) is equivalent to

(a) $d^2 = 0$ at $\boldsymbol{u} = 0$,
(b) $d\sigma^{(j)} + \sigma^{(j)}d = a_j \operatorname{id}$ at $\boldsymbol{u} = \boldsymbol{e}_j$,
(c) $(\sigma^{(j)})^2 = 0$ at $\boldsymbol{u} = 2\boldsymbol{e}_j$ and $\sigma^{(j)}\sigma^{(k)} + \sigma^{(k)}\sigma^{(j)} = 0$ at $\boldsymbol{u} = \boldsymbol{e}_j + \boldsymbol{e}_k$,

while at $|\boldsymbol{u}| \geq 3$ every term has a factor $\sigma_{\boldsymbol{v}}$ with $|\boldsymbol{v}| \geq 2$ and so vanishes. We call $|\boldsymbol{u}|$ a *level*, and refer to the largest $|\boldsymbol{u}|$ that appears in a system of higher homotopies as its level.

Remark 2.3. A system of higher homotopies on a complex of finite length must be of finite level, because $\sigma_{\boldsymbol{u}}$, which raises homological degree by $2|\boldsymbol{u}| - 1$, must eventually be a map into the 0 module. For example, on any complex of length $\ell \leq 2$ every system of higher homotopies is truncated.

We define a free resolution of R/I over $\overline{R} = R/(a_1, \dots, a_c)$ by finding a free resolution $F_\bullet$ of R/I over R, writing a copy of it for each monomial in c-variables, and gluing them together carefully. The t_j from the $R[t_j]_{j=1}^c$ behave as a shift of homological degree -2 on each of these free resolutions.

Definition 2.4. [29, Construction 2.4] For a system of higher homotopies $\sigma = \{\sigma_{\boldsymbol{u}}\}_{\boldsymbol{u}\in\mathbb{N}^c}$ for $a_1,\dots,a_c$ on $F_\bullet$, the *Eisenbud-Shamash resolution* is

$$\mathrm{Sh}(F_\bullet,\sigma) = R\{y_j\}_{j=1}^c \otimes_R F_\bullet \otimes_R \overline{R}, \qquad d_{\mathrm{Sh}} = \sum_{\boldsymbol{u}\in\mathbb{N}^c} t^{\boldsymbol{u}} \otimes \sigma_{\boldsymbol{u}} \otimes \mathrm{id}\,.$$

We write

$$D = \sum_{\boldsymbol{u}\in\mathbb{N}^c} t^{\boldsymbol{u}} \otimes \sigma_{\boldsymbol{u}} \colon R\{y_j\}_{j=1}^c \otimes_R F_\bullet \to R\{y_j\}_{j=1}^c \otimes_R F_\bullet,$$

so that $d_{\mathrm{Sh}} = \overline{D}$, in the following notation. A bar means reduction modulo $\mathfrak{a}$, that is, $-\otimes_R \overline{R}$. So $\overline{F}_\bullet = F_\bullet \otimes_R \overline{R}$ is free over $\overline{R}$ with basis $\bar\alpha = \alpha\otimes\overline{1}$ for $\alpha\in\mathcal{B}$, and an R-linear map g has reduction $\bar g = g\otimes\mathrm{id}$. Then $\mathrm{Sh}(F_\bullet,\sigma) = R\{y_j\}_{j=1}^c \otimes_R \overline{F}_\bullet$, with basis the $y^{\boldsymbol{u}}\otimes\bar\alpha$, and $d_{\mathrm{Sh}} = \overline{D} = \sum_{\boldsymbol{u}} t^{\boldsymbol{u}}\otimes\bar\sigma_{\boldsymbol{u}}$. The map D over R describes the differential of a curved complex [24, Chapter 3], while $\bar D$ is genuinely a differential over $\bar R$.

Remark 2.5. We will denote D^2 by Θ, and this will later be an important shorthand for computation in our proofs in Section 9.

We will provide a block-structure for this matrix in Section 4.

Although we call this resolution the Eisenbud-Shamash resolution, we will denote the differential d_{Sh}, the Shamash differential. It might not be apparent that it is a resolution. This fact will follow from the following due to Shamash [26] for $c = 1$ and Eisenbud [16, Theorem 7.2] in general. We have stated as it appears in [17, Proposition 4.1.4] and [29, Theorem 2.5].

Theorem 2.6 ([16, 17, 26, 29])**.** *Let $F_\bullet$ be a free resolution of R/I over R, let $a_1,\dots,a_c$ be an R-regular sequence in I, and let σ be a system of higher homotopies for $a_1,\dots,a_c$ on $F_\bullet$. Then $\mathrm{Sh}(F_\bullet,\sigma)$ is a free $\overline{R}$-resolution of R/I.*

Remark 2.7. The sum defining d_{Sh} is finite on each element, because $t^{\boldsymbol{u}}$ kills $y^{\boldsymbol{v}}$ unless $\boldsymbol{u}\le\boldsymbol{v}$. So d_{Sh} is well defined for every system of higher homotopies.

Remark 2.8. For a truncated system $\sigma = \{\sigma^{(j)}\}_{j=1}^c$ satisfying (a)-(c) in Definition 2.2 only $\boldsymbol{u} = \mathbf{0}$ and $\boldsymbol{u} = \boldsymbol{e}_j$ contribute, and

$$d_{\mathrm{Sh}} = \mathrm{id}\otimes d\otimes\mathrm{id} + \sum_{j=1}^c t_j\otimes\sigma^{(j)}\otimes\mathrm{id}\,.$$

3. Decomposing the Shamash differential

For a truncated system of higher homotopies, the differential of the Eisenbud-Shamash resolution has $c+1$ pieces, one for the base and one for each color. This section extends Remark 2.8, which will help make sense of the combinatorics of the Eisenbud-Shamash resolution.

Definition 3.1. For a truncated system $\sigma = \{\sigma^{(j)}\}_{j=1}^c$ satisfying (a)-(c) in Definition 2.2, we split D into the following maps.

- $\partial_0 = \mathrm{id}\otimes d$.
- $\partial_j = t_j\otimes\sigma^{(j)}$ for $1\le j\le c$, the *jth colored piece of the Shamash differential*. It lowers homological degree by 1, connecting the $\boldsymbol{u}$ copy of the base resolution in the Eisenbud-Shamash resolution to the $\boldsymbol{u}-\boldsymbol{e}_j$ copy.
- $\partial_{\mathrm{con}} = \partial_1+\cdots+\partial_c$, the *connecting part of the Shamash differential*.

Hence, $D = \partial_0+\partial_{\mathrm{con}}$ and $d_{\mathrm{Sh}} = \bar\partial_0+\bar\partial_{\mathrm{con}}$.

Remark 3.2. Since every t_j has even degree, no signs occur. Thus $\partial_j\partial_k = t_jt_k \otimes \sigma^{(j)}\sigma^{(k)}$ and $\partial_0\partial_j + \partial_j\partial_0 = t_j \otimes (d\sigma^{(j)} + \sigma^{(j)}d)$. The identities of a truncated system (Definition 2.2), therefore give on $R\{y_j\}_{j=1}^c \otimes_R F_\bullet$ before the tensor with $\overline{R}$,

$$\partial_0^2 = 0, \qquad \partial_{\mathrm{con}}^2 = \sum_j \partial_j^2 + \sum_{j<k}(\partial_j\partial_k + \partial_k\partial_j) = 0, \qquad \partial_0\partial_{\mathrm{con}} + \partial_{\mathrm{con}}\partial_0 = \sum_j t_j \otimes a_j \,\mathrm{id}\,.$$

Hence,

$$\Theta = D^2 = \partial_0^2 + (\partial_0\partial_{\mathrm{con}} + \partial_{\mathrm{con}}\partial_0) + \partial_{\mathrm{con}}^2 = \sum_{j=1}^c t_j \otimes a_j \,\mathrm{id},$$

and $d_{\mathrm{Sh}}^2 = \overline{D^2} = 0$ since each a_j is zero in $\overline{R}$. By this identity the CI operators t_j of $R\{y_j\}_{j=1}^c$ induce the maps $t_j \otimes \mathrm{id}$ which are the CI operators of $\mathrm{Sh}(F_\bullet, \sigma)$ for the lift D of d_{Sh}, in the sense of [17, Construction 4.1.1].

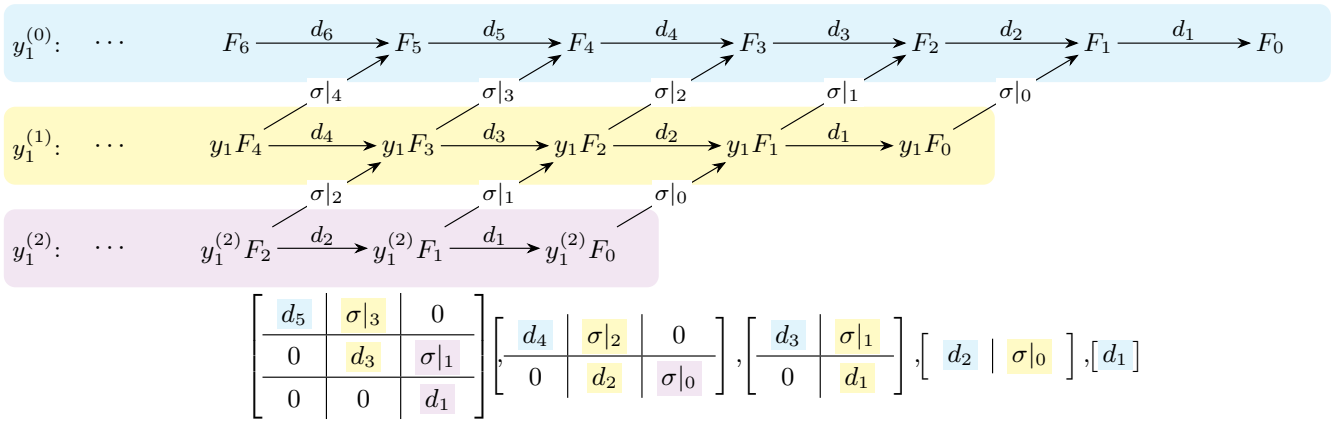


FIGURE 1. Shamash resolution of a truncated system, $c = 1$.

Figure 1 shows the Shamash resolution of a truncated system for $c = 1$, with $\sigma = \sigma^{(1)}$, the copy $y_1^{(i)}F_q$ in row i and column $n = 2i+q$, and $\bar\partial_0$ and $\bar\partial_1$ the horizontal and diagonal arrows. Below it are the matrices of $d_{\mathrm{Sh}}|_n$ for $n = 5, \ldots, 1$, with columns the copies in column n, rows those in column $n-1$, and $\sigma|_q$ the restriction of σ to F_q. Each matrix is block bidiagonal, corresponding to having a truncated system of higher homotopies. The general block structure has more superdiagonal terms because arrows don't stay between adjacent rows. We make this precise in Proposition 4.1.

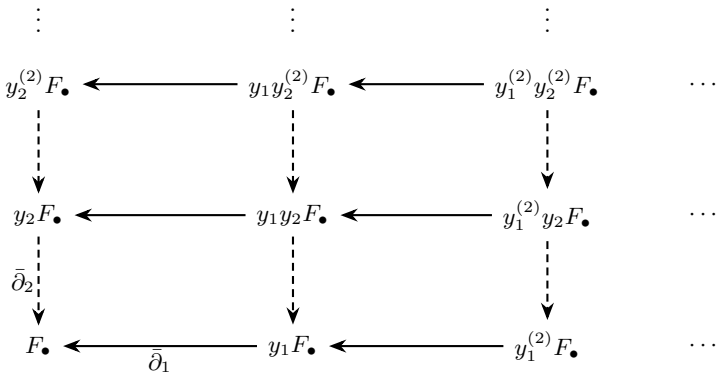


FIGURE 2. Shamash complex for $c = 2$.

Figure 2 illustrates a truncated system for $c = 2$. A general system adds an arrow from the copy over $\boldsymbol{u}$ to the copy over $\boldsymbol{u} - \boldsymbol{v}$ for every $\boldsymbol{v} \leq \boldsymbol{u}$ with $|\boldsymbol{v}| \geq 2$, so the lattice has additional diagonal arrows. The squares then no longer anticommute. Taking $\boldsymbol{u} = (1, 1)$ in (1) of Definition 2.2 gives $\sigma^{(1)}\sigma^{(2)} + \sigma^{(2)}\sigma^{(1)} = -(d\sigma_{(1,1)} + \sigma_{(1,1)}d)$, so

$$\partial_1\partial_2 + \partial_2\partial_1 = -t_1t_2 \otimes (d\sigma_{(1,1)} + \sigma_{(1,1)}d)$$

instead of zero, and the diagonal arrow $t_1t_2 \otimes \sigma_{(1,1)}$ accounts for the difference.

Solid arrows are $\bar{\partial}_1$, from the node at $\boldsymbol{u}$ to the node at $\boldsymbol{u} - \boldsymbol{e}_1$ of the slice behind, and dashed arrows are $\bar{\partial}_2$, to the node at $\boldsymbol{u} - \boldsymbol{e}_2$. The arrows of $\bar{\partial}_0$ go straight back from each node to the node at the same $\boldsymbol{u}$ and are not drawn. Every square anticommutes, because $\sigma^{(1)}\sigma^{(2)} + \sigma^{(2)}\sigma^{(1)} = 0$.

4. Block form of the Shamash differential

The Eisenbud-Shamash resolution takes a copy of a finite free resolution $F_\bullet$ for each lattice point $\boldsymbol{u} \in \mathbb{N}^c$, and groups them by level. Fix a level, k, and we should think of these lattice points as the points that are on the hyperplane $u_1 + \cdots + u_c = k$, k steps from the origin in the nonnegative orthant. The copy at each of these points is shifted up by $2k$ homological degrees, so consecutive hyperplanes are two homological degrees apart. In this section, we flatten this picture so that for each homological degree n and each hyperplane we determine which homological degree of the free resolutions that live there contributes to the degree n part of the Eisenbud-Shamash resolution. This allows us to make precise the block structure of the Shamash differential.

For $k \geq 0$ put $N_k = \#\{\boldsymbol{u} \in \mathbb{N}^c : |\boldsymbol{u}| = k\} = \binom{k+c-1}{c-1}$. Gathering summands by level $k = |\boldsymbol{u}|$,

$$(R\{y_j\}_{j=1}^c \otimes_R F_\bullet)_n = \bigoplus_{2k \leq n} L_k(n), \qquad L_k(n) = \bigoplus_{|\boldsymbol{u}|=k} y^{\boldsymbol{u}} \otimes F_{n-2k},$$

and $\mathrm{Sh}(F_\bullet, \sigma)_n = \bigoplus_{2k \leq n} L_k(n) \otimes_R \overline{R}$, with matrix entries read modulo $\mathfrak{a}$. The copy at a point $\boldsymbol{u}$ with $|\boldsymbol{u}| = k$ contributes $y^{\boldsymbol{u}} \otimes F_{n-2k}$ to the degree n part, as in Figure 3.

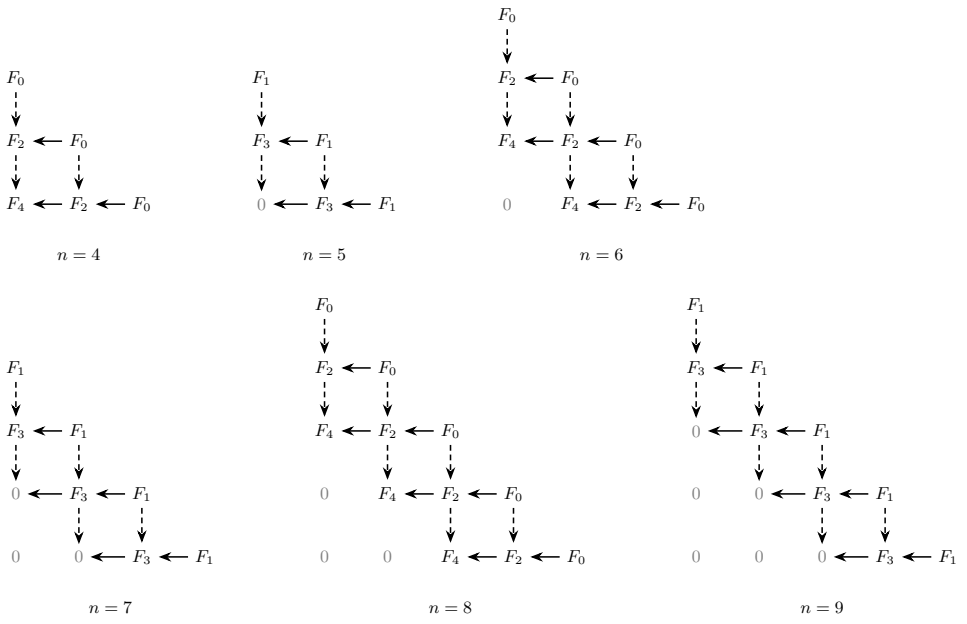
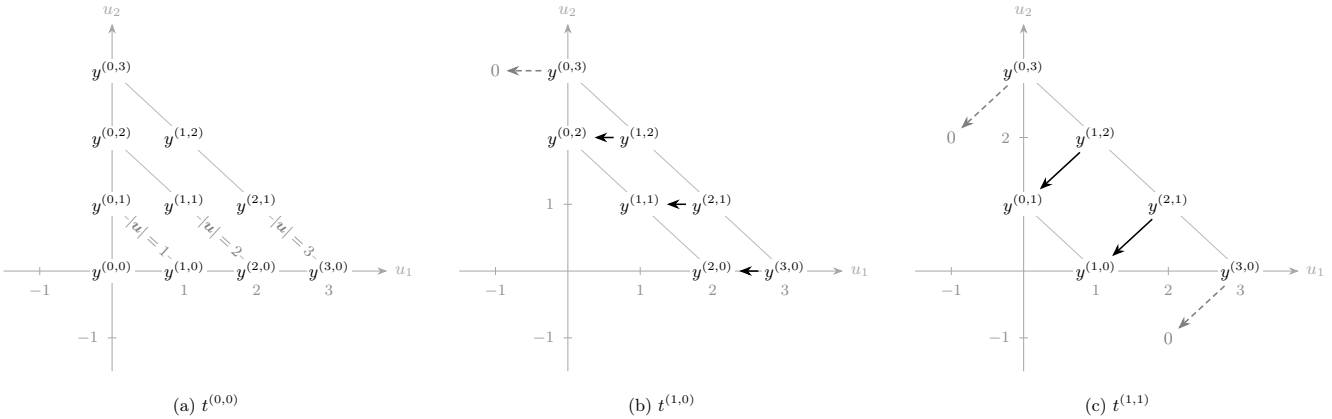


FIGURE 3. Degree n slices of the Shamash resolution of a truncated system, $c = 2$, $F_\bullet$ of length 4.

Let $\boldsymbol{v} \in \mathbb{N}^c$ and $k \geq |\boldsymbol{v}|$. The columns of $T_{\boldsymbol{v}}^{(k)}$ are indexed by the $y^{\boldsymbol{u}}$ with $|\boldsymbol{u}| = k$, in decreasing order of u_1, then of u_2, and so on, and its rows by the $y^{\boldsymbol{u}'}$ with $|\boldsymbol{u}'| = k - |\boldsymbol{v}|$, in the same order.

Its entry in column $y^{\boldsymbol{u}}$ and row $y^{\boldsymbol{u'}}$ is 1 if $\boldsymbol{u} = \boldsymbol{u'} + \boldsymbol{v}$ and 0 otherwise, so $T^{(k)}_{\boldsymbol{v}}$ is the matrix of $t^{\boldsymbol{v}}$ from level k to level $k - |\boldsymbol{v}|$, and $T^{(k)}_0 = \mathbf{1}_{N_k}$ is the identity. The columns are the points of the plane at k, and the rows are those of the plane at $k - |\boldsymbol{v}|$, and $t^{\boldsymbol{v}}$ translates by $-\boldsymbol{v}$, sending the points $\boldsymbol{u} \geq \boldsymbol{v}$ to $\boldsymbol{u} - \boldsymbol{v}$ and the others out of $\mathbb{N}^c$, to 0 (Figure 4).



(a) $t^{(0,0)}$ (b) $t^{(1,0)}$ (c) $t^{(1,1)}$

FIGURE 4. Sliding a plane by $t^{\boldsymbol{v}}$, $c = 2$.

For the plane $|\boldsymbol{u}| = 3$ of Figure 4, with columns $y^{(3,0)}$, $y^{(2,1)}$, $y^{(1,2)}$, $y^{(0,3)}$,

$$T^{(3)}_{(1,0)} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \qquad T^{(3)}_{(1,1)} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix},$$

the zero columns being the points carried out of $\mathbb{N}^2$. Since $t^{\boldsymbol{w}} t^{\boldsymbol{v}} = t^{\boldsymbol{v}+\boldsymbol{w}}$, these matrices compose as $T^{(k-|\boldsymbol{v}|)}_{\boldsymbol{w}} T^{(k)}_{\boldsymbol{v}} = T^{(k)}_{\boldsymbol{v}+\boldsymbol{w}}$. For an R-linear map f, $T \otimes f$ is the Kronecker product, the block matrix with block $T_{\boldsymbol{u},\boldsymbol{u'}} f$ in the column of $y^{\boldsymbol{u}}$ and the row of $y^{\boldsymbol{u'}}$, and $(T \otimes f)(T' \otimes g) = TT' \otimes fg$.

Proposition 4.1. *Let $F_\bullet$ be a free complex over R with a finite basis of each F_q, let $a_1, \ldots, a_c \in R$, and let $\sigma = \{\sigma_{\boldsymbol{u}}\}_{\boldsymbol{u} \in \mathbb{N}^c}$ be a system of higher homotopies for $a_1, \ldots, a_c$ on $F_\bullet$. Let $D = \sum_{\boldsymbol{u}} t^{\boldsymbol{u}} \otimes \sigma_{\boldsymbol{u}}$, acting on $R\{y_j\}_{j=1}^c \otimes_R F_\bullet$, so that $d_{\mathrm{Sh}} = \overline{D}$. Write its matrix from $(R\{y_j\}_{j=1}^c \otimes_R F_\bullet)_n$ to $(R\{y_j\}_{j=1}^c \otimes_R F_\bullet)_{n-1}$ in blocks, with rows indexed by the target level k' and columns by the source level k. Then:*

(1) The blocks are

$$\big(D|_n\big)_{k,k'} = \sum_{|\boldsymbol{v}| = k - k'} T^{(k)}_{\boldsymbol{v}} \otimes \sigma_{\boldsymbol{v}}|_{n-2k} \quad (k' \leq k), \qquad \big(D|_n\big)_{k,k'} = 0 \quad (k' > k),$$

so the matrix is block upper triangular and its mth superdiagonal carries exactly the $\sigma_{\boldsymbol{v}}$ with $|\boldsymbol{v}| = m$.

(2) σ is truncated if and only if the matrix of $D|_n$ is block bidiagonal for every n, in which case

$$\big(D|_n\big)_{k,k} = \mathbf{1}_{N_k} \otimes d_{n-2k}, \qquad \big(D|_n\big)_{k,k-1} = S_k|_{n-2k} \quad \textit{for} \quad S_k = \sum_{j=1}^{c} T^{(k)}_{\boldsymbol{e}_j} \otimes \sigma^{(j)}.$$

The matrix of the differential of $\mathrm{Sh}(F_\bullet, \sigma)$ is the same, with its entries read modulo $\mathfrak{a}$.

Proof. Part (1): For $|\boldsymbol{u}| = k$ and $\alpha \in F_{n-2k}$, the term $t^{\boldsymbol{v}} \otimes \sigma_{\boldsymbol{v}}$ sends $y^{\boldsymbol{u}} \otimes \alpha$ to $y^{\boldsymbol{u}-\boldsymbol{v}} \otimes \sigma_{\boldsymbol{v}}(\alpha) \in L_{k-|\boldsymbol{v}|}(n-1)$ if $\boldsymbol{v} \leq \boldsymbol{u}$ and to 0 otherwise. So its block from $y^{\boldsymbol{u}} \otimes F_{n-2k}$ to $y^{\boldsymbol{u'}} \otimes F_{n-1-2k'}$ is $\sigma_{\boldsymbol{v}}|_{n-2k}$ if $\boldsymbol{u} = \boldsymbol{u'} + \boldsymbol{v}$ and 0 otherwise, which is the $(\boldsymbol{u}, \boldsymbol{u'})$ block of $T^{(k)}_{\boldsymbol{v}} \otimes \sigma_{\boldsymbol{v}}|_{n-2k}$.

Part (2): If $\sigma_{\boldsymbol{v}} = 0$ for all $|\boldsymbol{v}| \geq 2$ then only $k' = k$ and $k' = k-1$ survive, and $T_0^{(k)} = \mathbf{1}_{N_k}$ gives the diagonal blocks.

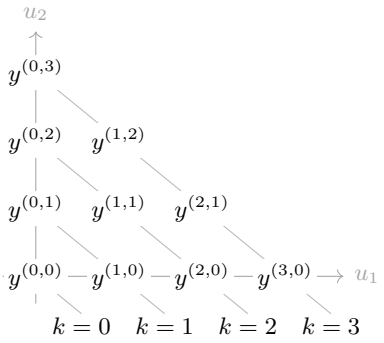
Conversely, if $\sigma_{\boldsymbol{v}}|_q \neq 0$ with $|\boldsymbol{v}| = m \geq 2$, take $n = 2m + q$. The block $(m, 0)$ is $\sum_{|\boldsymbol{v'}|=m} T_{\boldsymbol{v'}}^{(m)} \otimes \sigma_{\boldsymbol{v'}}|_q$, where $T_{\boldsymbol{v'}}^{(m)}$ is the $1 \times N_m$ matrix with its single 1 in column $\boldsymbol{v'}$, so the column of $\boldsymbol{v}$ carries $\sigma_{\boldsymbol{v}}|_q \neq 0$. □

By this proposition, we can write every block of an arbitrary Shamash differential down directly. We demonstrate this for $c = 2$. The (i, i') entry of $T_{(v_1,v_2)}^{(k)}$ is 1 exactly when $i' = i + v_2$, reading rows and columns from 0, so $T_{(v_1,v_2)}^{(k)}$ is a single band at offset v_2 and, for $0 \leq m \leq k$, the whole mth superdiagonal block, from the plane at level k to the plane at level $k - m$, is the banded Toeplitz array

$$\big(D|_n\big)_{k,k-m} = \left[\begin{array}{c|c|c|c|c|c|c} \sigma_{(m,0)} & \sigma_{(m-1,1)} & \sigma_{(m-2,2)} & \cdots & \sigma_{(0,m)} & & \\ \hline & \sigma_{(m,0)} & \sigma_{(m-1,1)} & \cdots & \sigma_{(1,m-1)} & \sigma_{(0,m)} & \\ \hline & & \ddots & \ddots & \ddots & \ddots & \ddots \end{array}\right]$$

of size $(k-m+1) \times (k+1)$, the blank entries being zero and every entry restricted to homological degree $n - 2k$.

At $n = 6$, with source levels $k = 0, 1, 2, 3$ and target levels $k' = 0, 1, 2$, the matrix of $D|_6$ is shown in Figure 5.



$D|_6 =$

	$k=0$	$k=1$	$k=2$	$k=3$
$k'=0$	d	$\left[\begin{array}{c\|c} \sigma_{(1,0)} & \sigma_{(0,1)} \end{array}\right]$	$\left[\begin{array}{c\|c\|c} \sigma_{(2,0)} & \sigma_{(1,1)} & \sigma_{(0,2)} \end{array}\right]$	$\left[\begin{array}{c\|c\|c\|c} \sigma_{(3,0)} & \sigma_{(2,1)} & \sigma_{(1,2)} & \sigma_{(0,3)} \end{array}\right]$
$k'=1$	0	$\left[\begin{array}{c\|c} d & 0 \\ \hline 0 & d \end{array}\right]$	$\left[\begin{array}{c\|c\|c} \sigma_{(1,0)} & \sigma_{(0,1)} & 0 \\ \hline 0 & \sigma_{(1,0)} & \sigma_{(0,1)} \end{array}\right]$	$\left[\begin{array}{c\|c\|c\|c} \sigma_{(2,0)} & \sigma_{(1,1)} & \sigma_{(0,2)} & 0 \\ \hline 0 & \sigma_{(2,0)} & \sigma_{(1,1)} & \sigma_{(0,2)} \end{array}\right]$
$k'=2$	0	0	$\left[\begin{array}{c\|c\|c} d & 0 & 0 \\ \hline 0 & d & 0 \\ \hline 0 & 0 & d \end{array}\right]$	$\left[\begin{array}{c\|c\|c\|c} \sigma_{(1,0)} & \sigma_{(0,1)} & 0 & 0 \\ \hline 0 & \sigma_{(1,0)} & \sigma_{(0,1)} & 0 \\ \hline 0 & 0 & \sigma_{(1,0)} & \sigma_{(0,1)} \end{array}\right]$

FIGURE 5. The planes at levels $0, 1, 2, 3$ for $c = 2$, and the matrix of $D|_6$, with the entries of column k restricted to degree $6 - 2k$.

For a truncated system, the block diagonal of Figure 5, the blocks $\mathbf{1}_{N_k} \otimes d$ from each level to itself, is $\partial_0 = \mathrm{id} \otimes d$, and the first block superdiagonal, the blocks from level k to level $k-1$, is $\partial_{\mathrm{con}} = \partial_1 + \cdots + \partial_c$. Each $\partial_j = t_j \otimes \sigma^{(j)}$ is the band $T_{\boldsymbol{e}_j}^{(k)} \otimes \sigma^{(j)}$ of these blocks. The blocks further

above the diagonal carry the $\sigma_{\boldsymbol{v}}$ with $|\boldsymbol{v}| \geq 2$, which vanish for a truncated system. So only the diagonal blocks and the connecting blocks survive,

$$S_k = \sum_{j=1}^{c} T^{(k)}_{\boldsymbol{e}_j} \otimes \sigma^{(j)}, \qquad S_2 = \begin{bmatrix} \sigma^{(1)} & \sigma^{(2)} & 0 \\ 0 & \sigma^{(1)} & \sigma^{(2)} \end{bmatrix} \quad (c = 2).$$

Remark 4.2. Counting the copies on each hyperplane,

$$\operatorname{rank} \operatorname{Sh}(F_\bullet, \sigma)_n = \sum_{2k \leq n} N_k \operatorname{rank} F_{n-2k}.$$

Remark 4.3 (The square of D)**.** Over R the square of D is not zero. With $\Theta = \sum_j t_j \otimes a_j \operatorname{id}$, the identity $D^2 = \partial_0\partial_{\mathrm{con}} + \partial_{\mathrm{con}}\partial_0 = \Theta$ of Remark 3.2 gives, for every n,

$$D|_n \, D|_{n+1} = \Theta|_{n+1}, \qquad \big(\Theta|_{n+1}\big)_{k,k-1} = \sum_{j=1}^{c} a_j T^{(k)}_{\boldsymbol{e}_j} \otimes \operatorname{id},$$

and every other block of $\Theta|_{n+1}$ is zero. So the product of two consecutive maps $D|_n$ is an explicit block matrix, with a_j in the positions where S_k has $\sigma^{(j)}$, one band for each color. This generalizes the identity $AB = BA = f \operatorname{id}$ of a matrix factorization to several elements $a_1, \ldots, a_c$ at once.

5. Truncated systems for DG resolutions

Truncation is a restriction on the system. On any DG resolution, and in particular on the Taylor resolution of a monomial ideal, a truncated system always exists.

Definition 5.1. [2, Section 1.3] Let A be a commutative ring. A *DG-algebra*, or differential graded algebra, over A is a complex $(F_\bullet, d)$ of finitely generated free A-modules with unity $1 \in F_0$ and associative multiplication that respects the homological grading and satisfies the following:

(1) $d(ab) = d(a)b + (-1)^{\operatorname{hdeg}(a)} d(b)$, for elements $a, b \in F_\bullet$, each in a single homological degree, and
(2) $ab = (-1)^{\operatorname{hdeg}(a)\operatorname{hdeg}(b)} ba$, for $a, b \in F_\bullet$.

When $F_\bullet$ is a free resolution which is also a DG-algebra, we call $F_\bullet$ a *DG-resolution*.

The following proposition is a standard result, and we include a short proof for completeness.

Proposition 5.2. *Let $F_\bullet$ be a DG-resolution of R/I with $F_0 = R$ and $x^2 = 0$ for every $x \in F_\bullet$ of odd homological degree, and let $a_1, \ldots, a_c \in I$. Since $a_i \in I = d(F_1)$, we may choose $z_i \in F_1$ with $dz_i = a_i$. Let $\sigma^{(i)}$ be multiplication by z_i, and $\sigma_{\boldsymbol{u}} = 0$ for $|\boldsymbol{u}| \geq 2$. Then σ is a truncated system of higher homotopies for $a_1, \ldots, a_c$ on $F_\bullet$.*

Proof. Since $d(z_i x) = a_i x - z_i \, d(x)$, we have $d\sigma^{(i)} + \sigma^{(i)} d = a_i \operatorname{id}$. Since $z_i^2 = 0$ and $z_i z_k = -z_k z_i$, we have $(\sigma^{(i)})^2 = 0$ and $\sigma^{(i)}\sigma^{(k)} + \sigma^{(k)}\sigma^{(i)} = 0$. These are the identities (a)-(c) of Definition 2.2. □

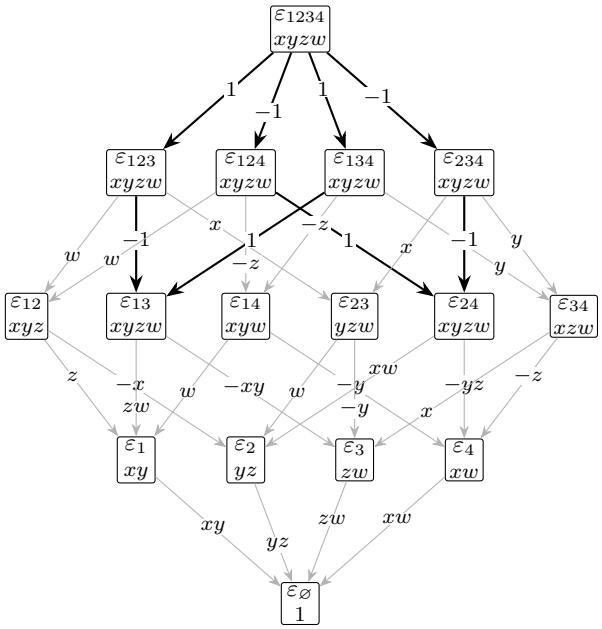
For a monomial ideal $I = (m_1, \ldots, m_r)$ the Taylor resolution is the free complex $F_\bullet$ whose cells ε_S are indexed by the subsets $S \subseteq \{1, \ldots, r\}$, where $\operatorname{hdeg}(\varepsilon_S) = |S|$ and $\operatorname{gr}(\varepsilon_S) = m_S = \operatorname{lcm}(m_i : i \in S)$, and whose differential is

$$d(\varepsilon_S) = \sum_{i \in S} (-1)^{\#\{s \in S : s > i\}} \frac{m_S}{m_{S \setminus i}} \varepsilon_{S \setminus i}.$$

The sign ensures that the first nonzero entry of every column in every matrix of d is positive. It counts the elements of S above the one being deleted, so the term that deletes the largest index carries a plus. This is the sign convention of [29, Def. 3.1], where the sign is written $(-1)^{k-p}$ at the pth element of $S = \{s_1 < \cdots < s_k\}$, with $k = |S|$, counting the same elements.

Ordering each basis $\{\varepsilon_S : |S| = q\}$ lexicographically, the row $S \setminus \max S$ is the first one the column of ε_S meets, and its entry is the one with the plus sign, so every column of every matrix of d has a positive leading entry. It is a free resolution of R/I with finitely many cells, multigraded for the fine grading. The Taylor resolution is a DG algebra [2], so Proposition 5.2 applies to it.

Example 5.3 (The running example)**.** Take $R = \Bbbk[x, y, z, w]$, the ideal $I = (xy, yz, zw, xw)$ with generators m_1, m_2, m_3, m_4 in that order, and $F_\bullet$ the Taylor resolution of R/I, with cells ε_S in homological degree $|S|$, ordered lexicographically. In Figure 6 we represent $G(F_\bullet)$, with the matrices d_q beside it.



$$d_4 = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$$

$$d_3 = \begin{bmatrix} w & w & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & -z & -z & 0 \\ x & 0 & 0 & x \\ 0 & 1 & 0 & -1 \\ 0 & 0 & y & y \end{bmatrix}$$

$$d_2 = \begin{bmatrix} z & zw & w & 0 & 0 & 0 \\ -x & 0 & 0 & w & xw & 0 \\ 0 & -xy & 0 & -y & 0 & x \\ 0 & 0 & -y & 0 & -yz & -z \end{bmatrix}$$

$$d_1 = \begin{bmatrix} xy & yz & zw & xw \end{bmatrix}$$

FIGURE 6. The running example: The Taylor resolution

In Figure 7, we illustrate a truncated system with

$$\sigma(\varepsilon_\varnothing) = \varepsilon_1, \qquad \sigma(\varepsilon_2) = -y\varepsilon_{12}, \quad \sigma(\varepsilon_3) = -\varepsilon_{13}, \quad \sigma(\varepsilon_4) = -x\varepsilon_{14},$$
$$\sigma(\varepsilon_{23}) = y\varepsilon_{123}, \quad \sigma(\varepsilon_{24}) = xy\varepsilon_{124}, \quad \sigma(\varepsilon_{34}) = x\varepsilon_{134}, \quad \sigma(\varepsilon_{234}) = -xy\varepsilon_{1234},$$

the dashed arrows of Figure 7.

On the Taylor resolution the homotopies can be written down, and the system they form is truncated. Sobieska gives the maps in [29, Def. 3.1] and the identities they satisfy in [29, Thm. 3.2], for any regular sequence in I and any choice of expressions for its members.

Theorem 5.4 ([29, Thm. 3.2])**.** *Let $I = (m_1, \dots, m_r)$ be a monomial ideal, let $F_\bullet$ be its Taylor resolution, and let $a_1, \dots, a_c$ be an R-regular sequence in I, not necessarily homogeneous. Fix expressions $a_j = \sum_{t=1}^{r} f_{j,t} m_t$ with $f_{j,t} \in R$. Write $S = \{s_1 < \cdots < s_k\}$ and let $p_{t,S}$ be the position of t in $S \cup t$, so that $S \cup t = \{s_1, \dots, s_{p_{t,S}-1}, t, s_{p_{t,S}}, \dots, s_k\}$. Then*

$$\sigma^{(j)}(\varepsilon_S) = \sum_{t \notin S} (-1)^{k - p_{t,S} - 1} f_{j,t} \frac{m_t m_S}{m_{S \cup t}} \varepsilon_{S \cup t}, \qquad \sigma_{\boldsymbol{u}} = 0 \text{ for } |\boldsymbol{u}| \geq 2,$$

is a system of higher homotopies for $a_1, \dots, a_c$ on $F_\bullet$. So every regular sequence in I has a truncated system on the Taylor resolution.

Since $m_{S\cup t} = \operatorname{lcm}(m_t, m_S)$ and $p_{t,S} - 1 = \#\{s \in S : s < t\}$, the maps read

$$\sigma^{(j)}(\varepsilon_S) = \sum_{t \notin S} (-1)^{\#\{s\in S : s>t\}} f_{j,t} \gcd(m_t, m_S) \varepsilon_{S\cup t},$$

with the sign convention of the Taylor differential. These are the maps of Proposition 5.2 for $z_j = \sum_{t=1}^{r} f_{j,t}\varepsilon_t$.

Proposition 5.5. *Let $I = (m_1, \dots, m_r)$ be a monomial ideal, $F_\bullet$ its Taylor resolution, and $a_1, \dots, a_c$ pairwise coprime monomials in I, so that they form a regular sequence. For each j pick a generator m_{g_j} dividing a_j, with $g_j \in \{1, \dots, r\}$, and use the one term expression $a_j = f_j m_{g_j}$, $f_j = a_j/m_{g_j}$, in Sobieska's construction.*

Then

$$\sigma^{(j)}(\varepsilon_S) = \pm \frac{a_j m_S}{m_{S\cup g_j}} \varepsilon_{S\cup g_j} \text{ for } g_j \notin S, \qquad \sigma^{(j)}(\varepsilon_S) = 0 \text{ for } g_j \in S,$$

so the $\sigma^{(j)}$ form a truncated system and each $\sigma^{(j)}$ is homogeneous of multidegree $\operatorname{gr}(a_j)$.

Moreover, if we label the cell $y^{\boldsymbol{u}} \otimes \varepsilon_S$ of $\operatorname{Sh}(F_\bullet, \sigma)$ by the monomial $a_1^{u_1} \cdots a_c^{u_c} m_S$, which is its multidegree, then every entry of d_{Sh} is, up to sign, the ratio of the labels of the two cells it joins.

Proof. Sobieska's sum in Theorem 5.4 has the single term $\pm f_j \gcd(m_{g_j}, m_S)\varepsilon_{S\cup g_j}$ when $g_j \notin S$ and none otherwise, and

$$f_j \gcd(m_{g_j}, m_S) = f_j m_{g_j} m_S / m_{S\cup g_j} = a_j m_S / m_{S\cup g_j}.$$

A level edge $y^{\boldsymbol{u}} \otimes \varepsilon_S \to y^{\boldsymbol{u}} \otimes \varepsilon_{S\setminus i}$ has weight $\pm m_S/m_{S\setminus i}$. A color-j connecting edge from $y^{\boldsymbol{u}} \otimes \varepsilon_S$ to $y^{\boldsymbol{u}-\boldsymbol{e}_j} \otimes \varepsilon_{S\cup g_j}$ has weight $\pm a_j m_S/m_{S\cup g_j}$. In each case, the weight is the ratio of the labels. □

Remark 5.6 (Monomial data)**.** Suppose I and $\mathfrak{a} = (a_1, \dots, a_c)$ are monomial ideals, $F_\bullet$ is multigraded in the fine grading with every nonzero $d_{\alpha,\beta}$ a scalar times a monomial, and each $\sigma^{(j)}$ is homogeneous of multidegree $\operatorname{gr}(a_j)$, as in the Taylor resolution with Sobieska's system. Give $y^{\boldsymbol{u}} \otimes \bar{\alpha}$ the multidegree $\sum_j u_j \operatorname{gr}(a_j) + \operatorname{gr}(\alpha)$. Then every nonzero entry of d_{Sh} is a scalar times

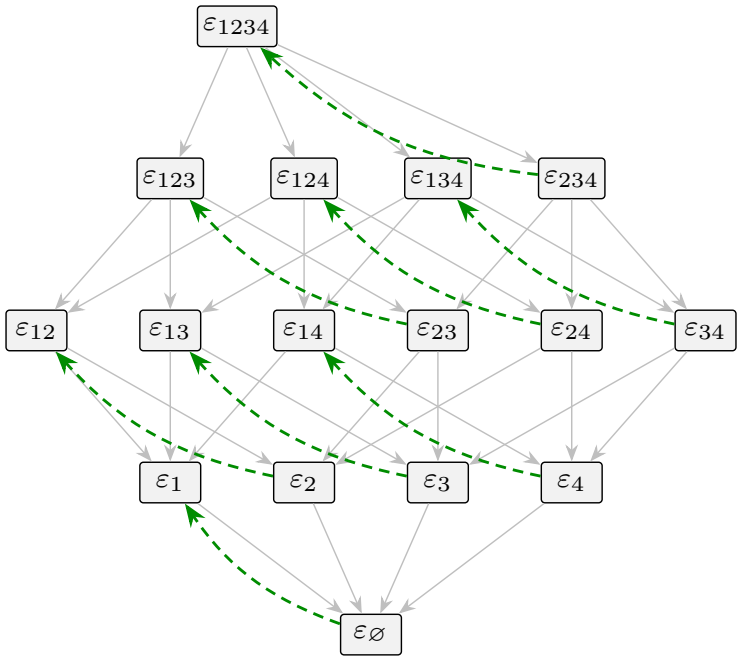


FIGURE 7. $G(F_\bullet)$ with the arrows of σ, $c = 1$.

the monomial of the difference of the multidegrees of its ends, so it is a unit exactly when its ends have equal multidegree.

So the Shamash resolution of a monomial ideal over a monomial complete intersection is combinatorial: its differential is determined by the labels, as for a cellular resolution, but several cells can carry the same label, so the copies of $F_\bullet$ over different $\boldsymbol{u}$ are kept apart. Proposition 5.5 and Remark 5.6 are the only places where the a_j must be monomials.

6. Algebraic Morse theory

This section fixes the form of algebraic Morse theory we use. For monomial ideals, Batzies and Welker [4] apply Forman's discrete Morse theory [19] to a CW complex supporting a cellular resolution. An acyclic matching of cells with equal monomial labels collapses the complex to a smaller CW complex, which again supports a cellular resolution, so the reduction respects cellularity. Algebraic Morse theory [23, 27] works with the free complex directly and needs no supporting cell complex. For monomial ideals there are algorithms that produce Morse matchings on the Taylor resolution: the pruning algorithm of Àlvarez Montaner, Fernández-Ramos and Gimenez [1, Algorithm 3.1], and the Barile–Macchia algorithm, introduced in [3] for edge ideals of forests and extended to all monomial ideals by Chau and Kara [11, Algorithm 2.9]. The results below apply to any of these matchings. Everything through Theorem 6.8 holds over an arbitrary commutative ring A, and we apply it over R and over $\overline{R}$.

Definition 6.1. Let A be a commutative ring. A *free complex* over A is a chain complex $(F_\bullet, d)$ of free A-modules together with a chosen basis $\mathcal{B}_q$ of each F_q. A basis element $\alpha \in \mathcal{B}_q$ is a *q-cell*, and $\mathcal{B} = \bigsqcup_q \mathcal{B}_q$.

The *entries* $d_{\alpha,\beta} \in A$ of d are defined by $d_q(\alpha) = \sum_{\beta \in \mathcal{B}_{q-1}} d_{\alpha,\beta}\beta$ for $\alpha \in \mathcal{B}_q$, a finite sum, and any collection of A-linear maps $f\colon F_\bullet \to F_{\bullet+r}$ has entries $f_{\alpha,\beta}$ defined the same way. The *complex graph* $G(F_\bullet)$ has vertex set $\mathcal{B}$ and an edge $\alpha \to \beta$ of *weight* $d_{\alpha,\beta}$ whenever $d_{\alpha,\beta} \neq 0$. An edge is a *unit edge* if its weight is a unit of A.

Remark 6.2. As seen from the definition, the basis is an important part of the data. Sköldberg calls the same object a *based complex* [27, §2].

Definition 6.3. A *partial matching* $\mathcal{M}$ on a free complex $F_\bullet$ over A is a set of unit edges of $G(F_\bullet)$, no two of which share a vertex. We use the following words for $\mathcal{M}$.

(1) For $(\alpha, \beta) \in \mathcal{M}$ we call α the *upper* and β the *lower* cell of the pair, the homological degree of β being one less than that of α. A cell lying on no edge of $\mathcal{M}$ is *critical*.
(2) The *replacement graph* $G^{\mathrm{rep}}(F_\bullet, \mathcal{M})$ has the vertices of $G(F_\bullet)$, the edges of $G(F_\bullet)$ that lie outside $\mathcal{M}$ with their weights unchanged, and for each $(\alpha, \beta) \in \mathcal{M}$ a reversed edge $\beta \to \alpha$ of weight $-(d_{\alpha,\beta})^{-1}$.
(3) A *directed path* in a directed graph is a sequence of edges, finite or infinite, each starting at the vertex where the previous one ends, with repeated vertices allowed.
(4) The matching is *acyclic* if $G^{\mathrm{rep}}(F_\bullet, \mathcal{M})$ contains no directed cycle, and *locally finite* if every directed path in $G^{\mathrm{rep}}(F_\bullet, \mathcal{M})$ is finite.
(5) A *Morse matching* is a locally finite partial matching.

Remark 6.4. Algebraic Morse theory is usually stated for *acyclic* matchings, those for which $G^{\mathrm{rep}}(F_\bullet, \mathcal{M})$ has no directed cycle [27, 23]. A locally finite matching is acyclic, since a directed cycle traversed over and over is an infinite directed path.

Given $\mathcal{M}$, split $d = \mu + \delta$, where the *matched part* μ keeps the entries $d_{\alpha,\beta}$ with $(\alpha, \beta) \in \mathcal{M}$ and the *unmatched part* is $\delta = d - \mu$.

Remark 6.5. For each homological degree q, we can partition the basis $\mathcal{B}_q$ into three sets K_q - the critical cells, U_q - the upper cells, and L_q - the lower cells. We can rewrite d_q as a block matrix with respect to the reordered bases $\{K_q, U_q, L_q\}$ and $\{K_{q-1}, U_{q-1}, L_{q-1}\}$ as illustrated below.

$$\begin{array}{c} \\ K_{q-1} \\ U_{q-1} \\ L_{q-1} \end{array} \begin{array}{c} \begin{array}{ccc} K_q & U_q & L_q \end{array} \\ \left[\begin{array}{c|c|c} d_{KK} & d_{UK} & d_{LK} \\ \hline d_{KU} & d_{UU} & d_{LU} \\ \hline d_{KL} & d_{UL} & d_{LL} \end{array}\right] \end{array},$$

where $d_{UL} = \delta_{UL} + \mu_{UL}$ with δ_{UL} representing the edges between upper and lower cells that are not matched edges, and μ_{UL} being all those edges that are in the matching. In terms of this ordering of the bases,

$$\delta = \left[\begin{array}{c|c|c} d_{KK} & d_{UK} & d_{LK} \\ \hline d_{KU} & d_{UU} & d_{LU} \\ \hline d_{KL} & \delta_{UL} & d_{LL} \end{array}\right] \text{ and } \mu = \left[\begin{array}{c|c|c} 0 & 0 & 0 \\ \hline 0 & 0 & 0 \\ \hline 0 & \mu_{UL} & 0 \end{array}\right],$$

where μ_{UL} is an invertible matrix by construction.

Definition 6.6. A *contraction* (i, p, h) of $(F_\bullet, d)$ onto $(F_\bullet', d')$ consists of chain maps $i\colon F_\bullet' \to F_\bullet$ and $p\colon F_\bullet \to F_\bullet'$ of homological degree 0 and a collection of A-linear map $h\colon F_\bullet \to F_{\bullet+1}$ of homological degree 1 with $pi = \mathrm{id}$, $ip - \mathrm{id} = dh + hd$ and $h^2 = hi = ph = 0$. In particular i and p are mutually inverse homotopy equivalences.

A *perturbation* of a complex $(F_\bullet, d)$ is an A-linear map $\delta\colon F_\bullet \to F_{\bullet-1}$ with $(d+\delta)^2 = 0$. For A-linear maps h and δ of a module to itself such that, for every element x, $(h\delta)^n x = 0$ for all large n, we write

$$\Phi = \sum_{n \geq 0} (h\delta)^n = (\mathrm{id} - h\delta)^{-1},$$

a finite sum on each element. Each time Φ appears we say which h and δ it is formed from. We state the perturbation lemma as Sköldberg does in [28, Theorem 1], except that his filtration serves only to make Φ a finite sum on each element, and we assume that directly.

Lemma 6.7 (Perturbation lemma, cf. [28, Theorem 1])**.** *Let (i, p, h) be a contraction of a complex $(F_\bullet, d)$ of A-modules onto a complex $(F_\bullet', d')$ of A-modules and let δ be a perturbation of $(F_\bullet, d)$ for which $\Phi = \sum_{n\geq 0}(h\delta)^n$ is a finite sum on each element. Then*

$$\big(\Phi i,\ p + p\delta\Phi h,\ \Phi h\big)$$

is a contraction of $(F_\bullet, d+\delta)$ onto $(F_\bullet', d' + p\delta\Phi i)$.

The main theorem of algebraic Morse theory is due independently to Sköldberg [27] and to Jöllenbeck and Welker [23, Theorem 2.2]. Sköldberg later showed that algebraic Morse theory follows from the perturbation lemma [28, Theorem 2], and the formulas for the contraction below are the ones his proof produces.

A *gradient path* π from α to β is a directed path $\alpha = v_0 \to v_1 \to \cdots \to v_{2m+1} = \beta$ in $G^{\mathrm{rep}}(F_\bullet, \mathcal{M})$, $m \geq 0$, in which $v_{2i} \to v_{2i+1}$ is an edge of $G(F_\bullet)$ outside $\mathcal{M}$ and $v_{2i+1} \to v_{2i+2}$ is a reversed matched edge. Its weight $\mathrm{wt}(\pi)$ is the product of the weights of its edges. A gradient path lowers homological degree by one, and for cells α and β with $\mathrm{hdeg}(\beta) = \mathrm{hdeg}(\alpha) - 1$ we write $\Gamma(\alpha, \beta) = \sum_\pi \mathrm{wt}(\pi)$, the sum over all gradient paths from α to β. For a Morse matching it has only finitely many nonzero terms by Theorem 6.8.

Theorem 6.8 (Algebraic Morse theory, [27, 23])**.** *Let $\mathcal{M}$ be a Morse matching on a free complex $(F_\bullet, d)$ over a commutative ring A, with $d = \mu + \delta$. Let $K_\bullet$ be the graded free A-module on the critical cells, $i\colon K_\bullet \to F_\bullet$ the inclusion of the critical cells and $p\colon F_\bullet \to K_\bullet$ the projection killing*

the noncritical cells. The map μ restricts to an isomorphism from the span of the upper cells onto the span of the lower cells, and we let h be minus its inverse on the lower cells and zero on the critical and upper cells.

Then (i, p, h) is a contraction of $(F_\bullet, \mu)$ onto $(K_\bullet, 0)$, the sum $\Phi = \sum_{n\geq 0}(h\delta)^n$ is finite on each element, and perturbing by δ gives a contraction of $(F_\bullet, d)$ onto the Morse complex $F_\bullet{}^{\mathcal{M}} = (K_\bullet, d^{\mathcal{M}})$,

$$\begin{array}{ccc} h \circlearrowright (F_\bullet, \mu) & \underset{i}{\overset{p}{\rightleftarrows}} & (K_\bullet, 0) \\ \Big\downarrow {\scriptstyle +\delta} & & \\ h^{\mathcal{M}} \circlearrowright (F_\bullet, d) & \underset{i^{\mathcal{M}}}{\overset{p^{\mathcal{M}}}{\rightleftarrows}} & F_\bullet{}^{\mathcal{M}} = (K_\bullet, d^{\mathcal{M}}) \end{array}$$

where

$$d^{\mathcal{M}} = p\delta\Phi i, \qquad d^{\mathcal{M}}_{\alpha,\beta} = \Gamma(\alpha, \beta) \quad \textit{for critical } \alpha, \beta, \qquad (i^{\mathcal{M}}, p^{\mathcal{M}}, h^{\mathcal{M}}) = \big(\Phi i,\ p + p\delta\Phi h,\ \Phi h\big).$$

Theorem 6.8 is Lemma 6.7 applied to (i, p, h) and the perturbation δ, since $(\mu + \delta)^2 = d^2 = 0$, with local finiteness making Φ a finite sum on each cell. It uses only that the matched entries are invertible. In the block form of Remark 6.5, with columns K_q, U_q, L_q, rows $K_{q+1}, U_{q+1}, L_{q+1}$, and μ_{UL} the matched block of d_{q+1},

$$h|_q = \left[\begin{array}{c|c|c} 0 & 0 & 0 \\ \hline 0 & 0 & -\mu_{UL}^{-1} \\ \hline 0 & 0 & 0 \end{array}\right].$$

Hence, when every matched weight is ± 1, as for the Taylor resolution, $h|_q = -\mu_{q+1}^T$.

In the graph, δ follows an edge outside $\mathcal{M}$, and h carries a lower cell up its reversed matched edge, with weight $-(d_{\alpha,\beta})^{-1}$, and kills every other cell. So $d^{\mathcal{M}} = p\delta\Phi i$ sums over gradient paths between critical cells. Likewise $i^{\mathcal{M}}$ and $p^{\mathcal{M}}$ sum over paths that alternate edges outside $\mathcal{M}$ and reversed matched edges, starting with an edge outside $\mathcal{M}$ for $i^{\mathcal{M}}$ and with a reversed edge for $p^{\mathcal{M}}$. So $i^{\mathcal{M}}(\kappa)$ involves only κ and upper cells, and $p^{\mathcal{M}}$ agrees with p except on the lower cells.

Example 6.9 (The running example)**.** We continue Example 5.3 with $R = \Bbbk[x, y, z, w]$, $I = (m_1, m_2, m_3, m_4) = (xy, yz, zw, xw)$ and the matching:

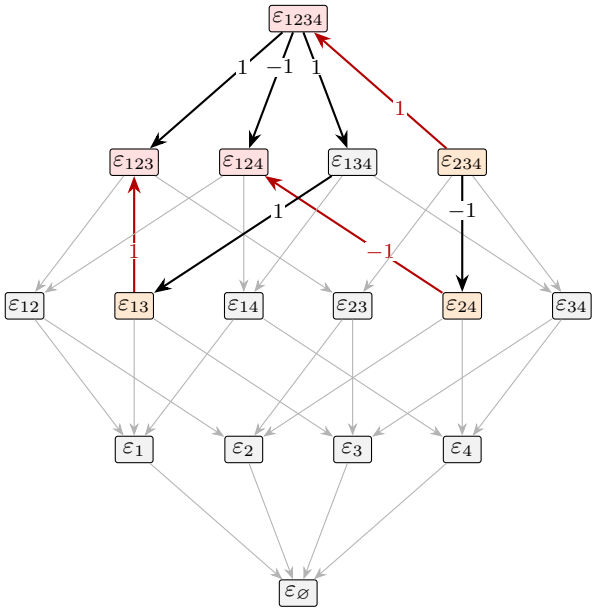
$$\mathcal{M} = \big\{(\varepsilon_{124}, \varepsilon_{24}), (\varepsilon_{123}, \varepsilon_{13}), (\varepsilon_{1234}, \varepsilon_{234})\big\}.$$

In Figure 8 we write $G^{\text{rep}}(F_\bullet, \mathcal{M})$ along with the maps μ and δ. $\mathcal{M}$ has no directed cycle, so $\mathcal{M}$ is a Morse matching. It is the matching produced by the pruning algorithm [1, Algorithm 3.1].

No entry of δ joins an upper cell to a lower cell, so $h^{\mathcal{M}} = h = -\mu^T$, which is zero except at the lower cells,

$$h(\varepsilon_{13}) = \varepsilon_{123}, \qquad h(\varepsilon_{24}) = -\varepsilon_{124}, \qquad h(\varepsilon_{234}) = \varepsilon_{1234}.$$

The map $p^{\mathcal{M}}$ differs from p only on the lower cells, for example $p^{\mathcal{M}}(\varepsilon_{13}) = w\varepsilon_{12} + x\varepsilon_{23}$ from the paths $\varepsilon_{13} \xrightarrow{1} \varepsilon_{123} \xrightarrow{w} \varepsilon_{12}$ and $\varepsilon_{13} \xrightarrow{1} \varepsilon_{123} \xrightarrow{x} \varepsilon_{23}$. In degrees 0 and 1 every cell is critical, i, p, $i^{\mathcal{M}}$, $p^{\mathcal{M}}$ are identities and $h = h^{\mathcal{M}} = 0$. In degrees 2 and 3, with the critical cells $\varepsilon_{12}, \varepsilon_{14}, \varepsilon_{23}, \varepsilon_{34}$ and ε_{134},



$$\mu_4 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ -1 \end{bmatrix}, \quad \delta_4 = \begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \end{bmatrix}$$

$$\mu_3 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \delta_3 = \begin{bmatrix} w & w & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -z & -z & 0 \\ x & 0 & 0 & x \\ 0 & 0 & 0 & -1 \\ 0 & 0 & y & y \end{bmatrix}$$

$$\mu_2 = 0, \quad \delta_2 = d_2$$

$$\mu_1 = 0, \quad \delta_1 = d_1$$

FIGURE 8. The running example. Red, orange, gray: upper, lower, critical cells.

the contraction (i, p, h) is

$$i|_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad p|_2 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad i|_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad p|_3 = \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix},$$

$$h|_2 = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad h|_3 = \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix},$$

and the contraction $(i^{\mathcal{M}}, p^{\mathcal{M}}, h^{\mathcal{M}})$ is

$$i^{\mathcal{M}}|_2 = i|_2, \quad i^{\mathcal{M}}|_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad p^{\mathcal{M}}|_2 = \begin{bmatrix} 1 & w & 0 & 0 & -w & 0 \\ 0 & 0 & 1 & 0 & z & 0 \\ 0 & x & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad p^{\mathcal{M}}|_3 = \begin{bmatrix} 0 & 0 & 1 & 1 \end{bmatrix}, \quad h^{\mathcal{M}} = h.$$

The Morse complex $F_\bullet{}^{\mathcal{M}}$ is the minimal free resolution of R/I with differentials

$$d_1^{\mathcal{M}} = d_1, \qquad d_2^{\mathcal{M}} = \begin{bmatrix} z & w & 0 & 0 \\ -x & 0 & w & 0 \\ 0 & 0 & -y & x \\ 0 & -y & 0 & -z \end{bmatrix}, \qquad d_3^{\mathcal{M}} = \begin{bmatrix} w \\ -z \\ x \\ y \end{bmatrix}.$$

Remark 6.10 (Row and column operations)**.** Algebraic Morse Theory can really be viewed as a combinatorial solution to the linear algebra problem of isolating the units in the matrices of a free resolution. The contraction $(i^{\mathcal{M}}, p^{\mathcal{M}}, h^{\mathcal{M}})$ of Theorem 6.8 gives a change of basis that splits $F_\bullet$ into $F_\bullet{}^{\mathcal{M}}$ and trivial complexes. Split $\mathcal{B}_q$ into the critical cells K_q, the upper matched cells

U_q and the lower matched cells L_q, and let ι_{U_q} be the inclusion of the span of U_q. The identity $i^{\mathcal{M}}p^{\mathcal{M}} - \mathrm{id} = dh^{\mathcal{M}} + h^{\mathcal{M}}d$ lets each x in F_q be expressed by

$$x = i^{\mathcal{M}}\big(p^{\mathcal{M}}(x)\big) + \big(-h^{\mathcal{M}}d(x)\big) + d\big(-h^{\mathcal{M}}(x)\big),$$

with $p^{\mathcal{M}}(x)$ in the span of K_q, $-h^{\mathcal{M}}d(x)$ in the span of U_q and $-h^{\mathcal{M}}(x)$ in the span of U_{q+1}. So the new basis of F_q is $i^{\mathcal{M}}(\kappa)$ for $\kappa \in K_q$, then $\alpha \in U_q$, then $d(\alpha')$ for $\alpha' \in U_{q+1}$, and the change of basis matrix and its inverse are

$$V_q = \begin{bmatrix} i^{\mathcal{M}}|_q & \iota_{U_q} & d\,\iota_{U_{q+1}} \end{bmatrix}, \qquad V_q^{-1} = \begin{bmatrix} p^{\mathcal{M}}|_q \\ -h^{\mathcal{M}}d|_q \\ -h^{\mathcal{M}}|_q \end{bmatrix}.$$

The columns of $i^{\mathcal{M}}|_q$ are the column operations on d_q, and the rows of $p^{\mathcal{M}}|_q$ are the row operations on d_{q+1}. In the new bases

$$V_{q-1}^{-1}\, d_q\, V_q = \begin{bmatrix} p^{\mathcal{M}} d\, i^{\mathcal{M}} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & \mathbf{1} & 0 \end{bmatrix} = \begin{bmatrix} d_q^{\mathcal{M}} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & \mathbf{1} & 0 \end{bmatrix},$$

so $F_\bullet \cong F_\bullet{}^{\mathcal{M}} \oplus \bigoplus_{(\alpha,\beta)\in\mathcal{M}}\big(0 \to A\alpha \to A\,d(\alpha) \to 0\big)$. For a single pair (α,β) of weight w, $i^{\mathcal{M}}(\gamma) = \gamma - w^{-1}d_{\gamma,\beta}\,\alpha$ subtracts $w^{-1}d_{\gamma,\beta}$ times column α from column γ, $p^{\mathcal{M}}(\beta) = -w^{-1}\sum_\varepsilon d_{\alpha,\varepsilon}\,\varepsilon$ subtracts $w^{-1}d_{\alpha,\varepsilon}$ times row β from row ε, and $d^{\mathcal{M}}_{\gamma,\varepsilon} = d_{\gamma,\varepsilon} - d_{\gamma,\beta}\,w^{-1}d_{\alpha,\varepsilon}$ is the Schur complement.

7. The Shamash complex as an input for Morse theory

This section applies Theorem 6.8 to the Shamash resolution, lifts a matching on $F_\bullet$ to it, and defines the transferred maps $\sigma^{\mathcal{M}}$. Throughout, $\overline{r}$ is the image in $\overline{R}$ of $r \in R$.

The Shamash resolution $\mathrm{Sh}(F_\bullet,\sigma)$ is free over $\overline{R}$ on the cells $y^{\boldsymbol{u}}\otimes\bar\alpha$, for $\boldsymbol{u}\in\mathbb{N}^c$ and $\alpha\in\mathcal{B}$, the cell $y^{\boldsymbol{u}}\otimes\bar\alpha$ in homological degree $2|\boldsymbol{u}| + \mathrm{hdeg}(\alpha)$. So it is zero in negative degrees and has finite rank $\sum_{2|\boldsymbol{u}|\le n}\operatorname{rank} F_{n-2|\boldsymbol{u}|}$ in degree n. On a cell

$$d_{\mathrm{Sh}}(y^{\boldsymbol{u}}\otimes\bar\alpha) = y^{\boldsymbol{u}}\otimes\bar d(\bar\alpha) + \sum_{j:u_j\ge 1} y^{\boldsymbol{u}-\boldsymbol{e}_j}\otimes\bar\sigma^{(j)}(\bar\alpha),$$

with coefficients read modulo $\mathfrak{a}$. In matrices, examples and figures we drop the bar and the tensor sign and write $y^{\boldsymbol{u}}\alpha$ for this cell, with α for $\boldsymbol{u}=\mathbf{0}$. So the complex graph of $\mathrm{Sh}(F_\bullet,\sigma)$ has two kinds of edges. A *level edge* $y^{\boldsymbol{u}}\otimes\bar\alpha \to y^{\boldsymbol{u}}\otimes\bar\beta$, of weight $\overline{d_{\alpha,\beta}}$, is an edge of $\bar\partial_0$, and a *color-j connecting edge* $y^{\boldsymbol{u}}\otimes\bar\alpha \to y^{\boldsymbol{u}-\boldsymbol{e}_j}\otimes\bar\beta$, of weight $\overline{\sigma^{(j)}_{\alpha,\beta}}$, is an edge of $\bar\partial_j$. A *connecting edge* is a color-j connecting edge for some j. The level edges over a fixed $\boldsymbol{u}$ are copies of edges of $G(F_\bullet)$, and since $\Bbbk\to\overline{R}$ is injective they include a copy of every unit edge of $G(F_\bullet)$.

Proposition 7.1. *Let $F_\bullet$ be a free resolution of R/I over R with a finite basis of each F_q, let $a_1,\dots,a_c$ be an R-regular sequence in I, let $\sigma = \{\sigma^{(j)}\}_{j=1}^c$ be a truncated system of higher homotopies for $a_1,\dots,a_c$ on $F_\bullet$, and let $\mathcal{A}$ be a Morse matching on $\mathrm{Sh}(F_\bullet,\sigma)$. Then the Morse complex $\mathrm{Sh}(F_\bullet,\sigma)^{\mathcal{A}}$, free over $\overline{R}$ on the cells critical for $\mathcal{A}$, is a free $\overline{R}$-resolution of R/I, and Theorem 6.8 gives a contraction of $\mathrm{Sh}(F_\bullet,\sigma)$ onto it.*

Proof. By Theorem 2.6, $\mathrm{Sh}(F_\bullet,\sigma)$ is a free $\overline{R}$-resolution of R/I. It is zero in negative degrees and of finite rank in each degree, so Theorem 6.8 applies over $\overline{R}$. A contraction is a homotopy equivalence, so $\mathrm{Sh}(F_\bullet,\sigma)^{\mathcal{A}}$ has the homology of $\mathrm{Sh}(F_\bullet,\sigma)$, which is R/I in degree 0. □

Lifting a matching. Given a partial matching $\mathcal{M}$ on a free resolution $F_\bullet$ of R/I, we can also produce a matching on the Eisenbud-Shamash complex by lifting it to $\mathrm{Sh}(F_\bullet, \sigma)$.

Definition 7.2. Let $F_\bullet$ be a free resolution of R/I over R with a finite basis of each F_q, let $a_1, \dots, a_c$ be an R-regular sequence in I, let $\sigma = \{\sigma^{(j)}\}_{j=1}^c$ be a truncated system of higher homotopies for $a_1, \dots, a_c$ on $F_\bullet$, and let $\mathcal{M}$ be a partial matching on $F_\bullet$. The *lift* of $\mathcal{M}$ to $\mathrm{Sh}(F_\bullet, \sigma)$ is

$$\mathscr{M} = \big\{(y^{\boldsymbol{u}} \otimes \bar{\alpha}, y^{\boldsymbol{u}} \otimes \bar{\beta}) : (\alpha, \beta) \in \mathcal{M}, \boldsymbol{u} \in \mathbb{N}^c\big\},$$

one copy of $\mathcal{M}$ over each $\boldsymbol{u}$. A cell $y^{\boldsymbol{u}} \otimes \bar{\alpha}$ is critical for $\mathscr{M}$ if and only if α is critical for $\mathcal{M}$, so the critical cells of $\mathscr{M}$ are the $y^{\boldsymbol{u}} \otimes \bar{\kappa}$ with κ critical for $\mathcal{M}$.

Proposition 7.3. *In the setting of Definition 7.2, if $\mathcal{M}$ is a Morse matching on $F_\bullet$ then $\mathscr{M}$ is a Morse matching on* $\mathrm{Sh}(F_\bullet, \sigma)$.

Proof. Pairs of $\mathscr{M}$ over distinct $\boldsymbol{u}$ are disjoint, and over the same $\boldsymbol{u}$ they come from disjoint pairs of $\mathcal{M}$. Each lies over a matched edge of $G(F_\bullet)$, whose weight is a nonzero scalar and stays a unit in $\overline{R}$. So $\mathscr{M}$ is a partial matching.

Every edge of $G^{\mathrm{rep}}(\mathrm{Sh}(F_\bullet, \sigma), \mathscr{M})$ is either a level or reversed edge, which preserves $\boldsymbol{u}$, or a connecting edge, which lowers $|\boldsymbol{u}|$ by one and is never reversed. Over a fixed $\boldsymbol{u}$, $y^{\boldsymbol{u}} \otimes \bar{\alpha} \mapsto \alpha$ carries edges injectively to edges of $G^{\mathrm{rep}}(F_\bullet, \mathcal{M})$, since the level edges over $\boldsymbol{u}$ are copies of edges of $G(F_\bullet)$. So a directed path from $y^{\boldsymbol{u}} \otimes \bar{\alpha}$ uses at most $|\boldsymbol{u}|$ connecting edges, and between them projects to directed paths of $G^{\mathrm{rep}}(F_\bullet, \mathcal{M})$, which are finite. Hence $\mathscr{M}$ is locally finite. □

Remark 7.4. For monomial data as in Remark 5.6, a level edge is a unit edge exactly when it copies one of $G(F_\bullet)$, and a color-j connecting edge $y^{\boldsymbol{u}} \otimes \bar{\alpha} \to y^{\boldsymbol{u}-\boldsymbol{e}_j} \otimes \bar{\beta}$ is a unit edge exactly when $\mathrm{gr}(\beta) = \mathrm{gr}(\alpha) + \mathrm{gr}(a_j)$. The same holds for the Morse complex $\mathrm{Sh}(F_\bullet, \sigma)^{\mathcal{A}}$ of a Morse matching $\mathcal{A}$ on $\mathrm{Sh}(F_\bullet, \sigma)$, which is therefore minimal exactly when it has no unit edges. A lifted matching contains no connecting edge, so reducing along a unit connecting edge needs a matching that is not lifted. This is why Proposition 7.1 allows arbitrary Morse matchings.

Definition 7.5. In the setting of Definition 7.2, let $\mathcal{M}$ be a Morse matching on $F_\bullet$, with Morse complex ${F_\bullet}^{\mathcal{M}} = (K_\bullet, d^{\mathcal{M}})$ and contraction $(i^{\mathcal{M}}, p^{\mathcal{M}}, h^{\mathcal{M}})$ of Theorem 6.8. Set $\sigma_0^{\mathcal{M}} = d^{\mathcal{M}}$ and, for $|\boldsymbol{u}| = k \geq 1$,

$$\sigma_{\boldsymbol{u}}^{\mathcal{M}}|_q = \sum_{\substack{(j_1,\dots,j_k) \\ \boldsymbol{e}_{j_1}+\cdots+\boldsymbol{e}_{j_k}=\boldsymbol{u}}} p^{\mathcal{M}}|_{q+2k-1} \circ \sigma^{(j_1)}|_{q+2k-2} \circ h^{\mathcal{M}}|_{q+2k-3} \circ \sigma^{(j_2)}|_{q+2k-4} \circ \cdots \circ h^{\mathcal{M}}|_{q+1} \circ \sigma^{(j_k)}|_q \circ i^{\mathcal{M}}|_q, \tag{2}$$

for every q, that is,

$$\sigma_{\boldsymbol{u}}^{\mathcal{M}} = \sum p^{\mathcal{M}} \sigma^{(j_1)} h^{\mathcal{M}} \sigma^{(j_2)} \cdots h^{\mathcal{M}} \sigma^{(j_k)} i^{\mathcal{M}}$$

over the $\binom{k}{u_1,\dots,u_c}$ sequences in which the color j occurs u_j times. In particular $\sigma_{\boldsymbol{e}_j}^{\mathcal{M}} = p^{\mathcal{M}} \sigma^{(j)} i^{\mathcal{M}}$.

The formula (2) can be read off a picture like Figure 9, where the reversed matched edges of $G^{\mathrm{rep}}(F_\bullet, \mathcal{M})$ are red, its other edges are black (drawn in gray), and the arrows of the $\sigma^{(j)}$ are colored. Start at a critical cell κ. Follow a red-black path that stays in the homological degree of κ and the one below it and ends in the degree of κ, and then a colored edge of color j_k. From the cell this reaches, follow a red-black path that starts up a red edge and then a new colored edge of color j_{k-1}, and continue in this way through all k colored edges. After the last colored edge, follow a red-black path that eventually ends on a critical cell κ'. These three kinds of red-black paths are the terms of $i^{\mathcal{M}}$, $h^{\mathcal{M}}$ and $p^{\mathcal{M}}$. The coefficient of a critical cell κ' in $\sigma_{\boldsymbol{u}}^{\mathcal{M}}(\kappa)$ is the sum, over all such paths from κ to κ', of the product of the weights of the edges on the path.

Each summand raises homological degree by $2k-1$, the k factors $\sigma^{(j_r)}$ and the $k-1$ factors $h^{\mathcal{M}}$ each raising by one homological degree while $p^{\mathcal{M}}$ and $i^{\mathcal{M}}$ preserve the homological degree. So $\sigma_{\boldsymbol{u}}^{\mathcal{M}}$ raises homological degree by $2|\boldsymbol{u}| - 1$.

8. Motivating the Main Theorem

The examples in this section continue the running example 6.9 with Sobieska's homotopies from Section 5 to motivate and illustrate the Main Theorem 9.1 which we present in Section 9.

Example 8.1 (The running example, $c = 1$)**.** Take the running example, Example 6.9 with $c = 1$, $a_1 = m_1 = xy$ and σ Sobieska's homotopy (Theorem 5.4).

The nonzero values of σ on cells are

$$\sigma(\varepsilon_\varnothing) = \varepsilon_1, \qquad \sigma(\varepsilon_2) = -y\varepsilon_{12}, \quad \sigma(\varepsilon_3) = -\varepsilon_{13}, \quad \sigma(\varepsilon_4) = -x\varepsilon_{14},$$
$$\sigma(\varepsilon_{23}) = y\varepsilon_{123}, \quad \sigma(\varepsilon_{24}) = xy\varepsilon_{124}, \quad \sigma(\varepsilon_{34}) = x\varepsilon_{134}, \quad \sigma(\varepsilon_{234}) = -xy\varepsilon_{1234},$$

the dashed arrows of Figure 9.

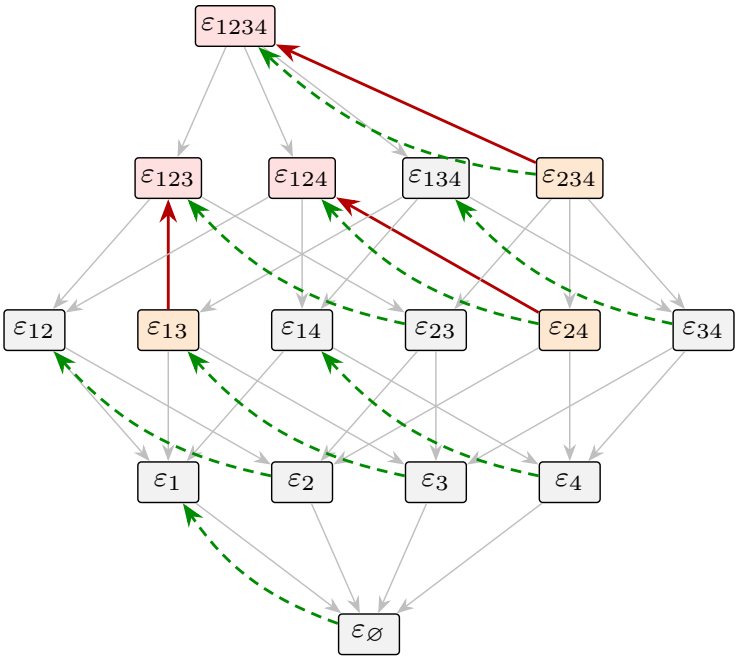


Figure 9. $G(F_\bullet)$ with the arrows of σ, $c = 1$.

In the bases of Section 6,

$$\sigma|_0 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \sigma|_1 = \begin{bmatrix} 0 & -y & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -x \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

$$\sigma|_2 = \begin{bmatrix} 0 & 0 & 0 & y & 0 & 0 \\ 0 & 0 & 0 & 0 & xy & 0 \\ 0 & 0 & 0 & 0 & 0 & x \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \sigma|_3 = \begin{bmatrix} 0 & 0 & 0 & -xy \end{bmatrix},$$

with $\sigma|_4 = 0$.

Over $R/(xy)$, $\mathrm{Sh}(F_\bullet, \sigma)$ has ranks $1, 4, 7, 8, 8, \ldots$. With cells ordered by level, d_{Sh} has the block form of Figure 1, with first matrices

$$d_{\mathrm{Sh}}|_1 = d_1 = \begin{matrix} \varepsilon_1 & \varepsilon_2 & \varepsilon_3 & \varepsilon_4 \end{matrix} \begin{bmatrix} 0 & yz & zw & xw \end{bmatrix} \varepsilon_\varnothing,$$

$$d_{\mathrm{Sh}}|_2 = \left[\begin{array}{c|c} d_2 & \sigma|_0 \end{array}\right] = \begin{array}{c} \begin{matrix} \varepsilon_{12} & \varepsilon_{13} & \varepsilon_{14} & \varepsilon_{23} & \varepsilon_{24} & \varepsilon_{34} \end{matrix} \\ \left[\begin{array}{cccccc|c} z & zw & w & 0 & 0 & 0 & 1 \\ -x & 0 & 0 & w & xw & 0 & 0 \\ 0 & 0 & 0 & -y & 0 & x & 0 \\ 0 & 0 & -y & 0 & -yz & -z & 0 \end{array}\right] \end{array} \begin{matrix} y_1\varepsilon_\varnothing \\ \\ \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \varepsilon_4, \end{matrix}$$

$$d_{\mathrm{Sh}}|_3 = \left[\begin{array}{c|c} d_3 & \sigma|_1 \\ \hline 0 & d_1 \end{array}\right] = \begin{array}{c} \begin{matrix} \varepsilon_{123} & \varepsilon_{124} & \varepsilon_{134} & \varepsilon_{234} & y_1\varepsilon_1 & y_1\varepsilon_2 & y_1\varepsilon_3 & y_1\varepsilon_4 \end{matrix} \\ \left[\begin{array}{cccc|cccc} w & w & 0 & 0 & 0 & -y & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 & 0 & -1 & 0 \\ 0 & -z & -z & 0 & 0 & 0 & 0 & -x \\ x & 0 & 0 & x & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & y & y & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & yz & zw & xw \end{array}\right] \end{array} \begin{matrix} \varepsilon_{12} \\ \varepsilon_{13} \\ \varepsilon_{14} \\ \varepsilon_{23} \\ \varepsilon_{24} \\ \varepsilon_{34} \\ \\ y_1\varepsilon_\varnothing, \end{matrix}$$

$$d_{\mathrm{Sh}}|_4 = \left[\begin{array}{c|c|c} d_4 & \sigma|_2 & 0 \\ \hline 0 & d_2 & \sigma|_0 \end{array}\right] = \begin{array}{c} \begin{matrix} \varepsilon_{1234} & y_1\varepsilon_{12} & y_1\varepsilon_{13} & y_1\varepsilon_{14} & y_1\varepsilon_{23} & y_1\varepsilon_{24} & y_1\varepsilon_{34} & y_1^{(2)}\varepsilon_\varnothing \end{matrix} \\ \left[\begin{array}{c|ccccccc} 1 & 0 & 0 & 0 & y & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & xy & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & x & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hline 0 & z & zw & w & 0 & 0 & 0 & 1 \\ 0 & -x & 0 & 0 & w & xw & 0 & 0 \\ 0 & 0 & 0 & 0 & -y & 0 & x & 0 \\ 0 & 0 & 0 & -y & 0 & -yz & -z & 0 \end{array}\right] \end{array} \begin{matrix} \varepsilon_{123} \\ \varepsilon_{124} \\ \varepsilon_{134} \\ \varepsilon_{234} \\ \\ y_1\varepsilon_1 \\ y_1\varepsilon_2 \\ y_1\varepsilon_3 \\ y_1\varepsilon_4, \end{matrix}$$

$$d_{\mathrm{Sh}}|_5 = \left[\begin{array}{c|c} \sigma|_3 & 0 \\ \hline d_3 & \sigma|_1 \\ \hline 0 & d_1 \end{array}\right] = \begin{array}{c} \begin{matrix} y_1\varepsilon_{123} & y_1\varepsilon_{124} & y_1\varepsilon_{134} & y_1\varepsilon_{234} & y_1^{(2)}\varepsilon_1 & y_1^{(2)}\varepsilon_2 & y_1^{(2)}\varepsilon_3 & y_1^{(2)}\varepsilon_4 \end{matrix} \\ \left[\begin{array}{cccc|cccc} 0 & 0 & 0 & -xy & 0 & 0 & 0 & 0 \\ \hline w & w & 0 & 0 & 0 & -y & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 & 0 & -1 & 0 \\ 0 & -z & -z & 0 & 0 & 0 & 0 & -x \\ x & 0 & 0 & x & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & y & y & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & yz & zw & xw \end{array}\right] \end{array} \begin{matrix} \varepsilon_{1234} \\ \\ y_1\varepsilon_{12} \\ y_1\varepsilon_{13} \\ y_1\varepsilon_{14} \\ y_1\varepsilon_{23} \\ y_1\varepsilon_{24} \\ y_1\varepsilon_{34} \\ \\ y_1^{(2)}\varepsilon_\varnothing. \end{matrix}$$

For $q \geq 2$, $d_{\mathrm{Sh}}|_{2q} = d_{\mathrm{Sh}}|_4$ and $d_{\mathrm{Sh}}|_{2q+1} = d_{\mathrm{Sh}}|_5$. The bases for the columns and rows for $d_{\mathrm{Sh}}|_{2q}$ are

columns: $y_1^{(q-2)}\varepsilon_{1234}$, $y_1^{(q-1)}\varepsilon_{12}$, $y_1^{(q-1)}\varepsilon_{13}$, $y_1^{(q-1)}\varepsilon_{14}$, $y_1^{(q-1)}\varepsilon_{23}$, $y_1^{(q-1)}\varepsilon_{24}$, $y_1^{(q-1)}\varepsilon_{34}$, $y_1^{(q)}\varepsilon_\varnothing$;

rows: $y_1^{(q-2)}\varepsilon_{123}$, $y_1^{(q-2)}\varepsilon_{124}$, $y_1^{(q-2)}\varepsilon_{134}$, $y_1^{(q-2)}\varepsilon_{234}$, $y_1^{(q-1)}\varepsilon_1$, $y_1^{(q-1)}\varepsilon_2$, $y_1^{(q-1)}\varepsilon_3$, $y_1^{(q-1)}\varepsilon_4$.

The bases for the columns and rows for $d_{\mathrm{Sh}}|_{2q+1}$ are

columns: $y_1^{(q-1)}\varepsilon_{123}$, $y_1^{(q-1)}\varepsilon_{124}$, $y_1^{(q-1)}\varepsilon_{134}$, $y_1^{(q-1)}\varepsilon_{234}$, $y_1^{(q)}\varepsilon_1$, $y_1^{(q)}\varepsilon_2$, $y_1^{(q)}\varepsilon_3$, $y_1^{(q)}\varepsilon_4$;

rows: $y_1^{(q-2)}\varepsilon_{1234}$, $y_1^{(q-1)}\varepsilon_{12}$, $y_1^{(q-1)}\varepsilon_{13}$, $y_1^{(q-1)}\varepsilon_{14}$, $y_1^{(q-1)}\varepsilon_{23}$, $y_1^{(q-1)}\varepsilon_{24}$, $y_1^{(q-1)}\varepsilon_{34}$, $y_1^{(q)}\varepsilon_\varnothing$.

The edges in our matching $\mathcal{M}$ are $(\varepsilon_{123},\varepsilon_{13})$, $(\varepsilon_{124},\varepsilon_{24})$ and $(\varepsilon_{1234},\varepsilon_{234})$. By the definition of $\mathcal{M}$, the edges of $\mathcal{M}$ are

$$\{(y_1^{(q)}\varepsilon_{123},y_1^{(q)}\varepsilon_{13}),(y_1^{(q)}\varepsilon_{124},y_1^{(q)}\varepsilon_{24}),(y_1^{(q)}\varepsilon_{1234},y_1^{(q)}\varepsilon_{234})\}_{q=0}^{\infty}.$$

There are also unit connecting edges $y_1^{(q+1)}\varepsilon_{\varnothing}\xrightarrow{1}y_1^{(q)}\varepsilon_1$ and $y_1^{(q+1)}\varepsilon_3\xrightarrow{-1}y_1^{(q)}\varepsilon_{13}$ for all $q\geq 0$. These are shown in Figure 10, where dashed red arrows point toward partners in degree 7. No lifted matching contains these edges.

To illustrate Theorem 9.1, note that the transferred system is given by the formula $\sigma^{\mathcal{M}}=p^{\mathcal{M}}\sigma i^{\mathcal{M}}$ of Definition 7.5, and the terms $\sigma_k^{\mathcal{M}}$ with $k\geq 2$ vanish since $p^{\mathcal{M}}\sigma(h^{\mathcal{M}}\sigma)^k i^{\mathcal{M}}=0$ for $k\geq 1$. Only $\sigma(\varepsilon_3)=-\varepsilon_{13}$ lands on a lower cell, so $\sigma^{\mathcal{M}}=p\sigma i$ except at ε_3, where the paths $\varepsilon_{13}\xrightarrow{1}\varepsilon_{123}\xrightarrow{w}\varepsilon_{12}$ and $\varepsilon_{13}\xrightarrow{1}\varepsilon_{123}\xrightarrow{x}\varepsilon_{23}$ give

$$\sigma^{\mathcal{M}}(\varepsilon_3)=-p^{\mathcal{M}}(\varepsilon_{13})=-w\varepsilon_{12}-x\varepsilon_{23},$$

so

$$\sigma^{\mathcal{M}}|_0=\begin{bmatrix}1\\0\\0\\0\end{bmatrix},\qquad \sigma^{\mathcal{M}}|_1=\begin{bmatrix}0&-y&-w&0\\0&0&0&-x\\0&0&-x&0\\0&0&0&0\end{bmatrix},\qquad \sigma^{\mathcal{M}}|_2=\begin{bmatrix}0&0&0&x\end{bmatrix}.$$

Note that the path $\varepsilon_3\to\varepsilon_{13}\to\varepsilon_{123}\to\varepsilon_{23}\to\varepsilon_{123}$ does not contribute to $\sigma^{\mathcal{M}}$ because a path described by $h^{\mathcal{M}}$ must start and end with a red arrow.

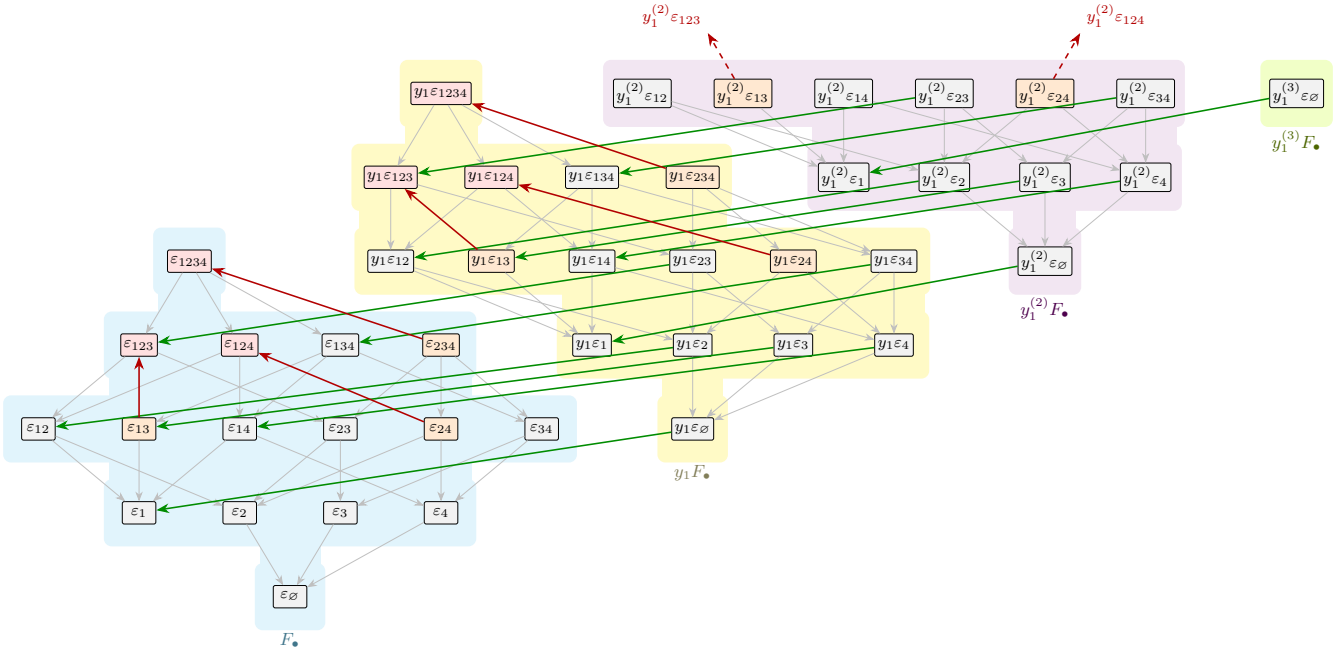


FIGURE 10. $G^{\mathrm{rep}}\big(\mathrm{Sh}(F_\bullet,\sigma),\mathcal{M}\big)$ through degree 6, $c=1$. Green: $\bar{\partial}_1$.

The reduced complex $\mathrm{Sh}(F_\bullet,\sigma)^{\mathcal{M}}$ over $\overline{R}=R/(xy)$ has ranks $1,4,5,5,5,\ldots$ and matrices

$$d_{\mathrm{Sh}}^{\mathcal{M}}|_1=\big[d_1^{\mathcal{M}}\big]=\begin{matrix}\varepsilon_1&\varepsilon_2&\varepsilon_3&\varepsilon_4\end{matrix}\\\begin{bmatrix}0&yz&zw&xw\end{bmatrix}\ \varepsilon_{\varnothing},$$

$$d^{\mathcal{M}}_{\mathrm{Sh}}|_2 = \left[\begin{array}{c|c|c|c|c} & \varepsilon_{12} & \varepsilon_{14} & \varepsilon_{23} & \varepsilon_{34} \\ d^{\mathcal{M}}_2 & z & w & 0 & 0 \\ & -x & 0 & w & 0 \\ & 0 & 0 & -y & x \\ & 0 & -y & 0 & -z \end{array}\right] \begin{array}{|c} y_1\varepsilon_\varnothing \\ 1 \\ 0 \\ 0 \\ 0 \end{array} \begin{array}{l} \\ \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \varepsilon_4, \end{array}$$

$$d^{\mathcal{M}}_{\mathrm{Sh}}|_3 = \left[\begin{array}{c|c} d^{\mathcal{M}}_3 & \sigma^{\mathcal{M}}|_1 \\ \hline 0 & d^{\mathcal{M}}_1 \end{array}\right] = \left[\begin{array}{c|cccc} \varepsilon_{134} & y_1\varepsilon_1 & y_1\varepsilon_2 & y_1\varepsilon_3 & y_1\varepsilon_4 \\ w & 0 & -y & -w & 0 \\ -z & 0 & 0 & 0 & -x \\ x & 0 & 0 & -x & 0 \\ y & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & yz & zw & xw \end{array}\right] \begin{array}{l} \varepsilon_{12} \\ \varepsilon_{14} \\ \varepsilon_{23} \\ \varepsilon_{34} \\ \hline y_1\varepsilon_\varnothing, \end{array}$$

$$d^{\mathcal{M}}_{\mathrm{Sh}}|_4 = \left[\begin{array}{c|c} \sigma^{\mathcal{M}}|_2 & 0 \\ \hline d^{\mathcal{M}}_2 & \sigma^{\mathcal{M}}|_0 \end{array}\right] = \left[\begin{array}{cccc|c} y_1\varepsilon_{12} & y_1\varepsilon_{14} & y_1\varepsilon_{23} & y_1\varepsilon_{34} & y_1^{(2)}\varepsilon_\varnothing \\ \hline 0 & 0 & 0 & x & 0 \\ z & w & 0 & 0 & 1 \\ -x & 0 & w & 0 & 0 \\ 0 & 0 & -y & x & 0 \\ 0 & -y & 0 & -z & 0 \end{array}\right] \begin{array}{l} \varepsilon_{134} \\ \hline y_1\varepsilon_1 \\ y_1\varepsilon_2 \\ y_1\varepsilon_3 \\ y_1\varepsilon_4. \end{array}$$

For all $q \geq 2$, $d^{\mathcal{M}}_{\mathrm{Sh}}|_{2q-1} = d^{\mathcal{M}}_{\mathrm{Sh}}|_3$ and $d^{\mathcal{M}}_{\mathrm{Sh}}|_{2q} = d^{\mathcal{M}}_{\mathrm{Sh}}|_4$. The bases for $d^{\mathcal{M}}_{\mathrm{Sh}}|_{2q-1}$ are

$$\text{columns } y_1^{(q-2)}\varepsilon_{134},\ y_1^{(q-1)}\varepsilon_1,\ y_1^{(q-1)}\varepsilon_2,\ y_1^{(q-1)}\varepsilon_3,\ y_1^{(q-1)}\varepsilon_4;$$
$$\text{rows } y_1^{(q-2)}\varepsilon_{12},\ y_1^{(q-2)}\varepsilon_{14},\ y_1^{(q-2)}\varepsilon_{23},\ y_1^{(q-2)}\varepsilon_{34},\ y_1^{(q-1)}\varepsilon_\varnothing.$$

The bases for $d^{\mathcal{M}}_{\mathrm{Sh}}|_{2q}$ are

$$\text{columns } y_1^{(q-1)}\varepsilon_{12},\ y_1^{(q-1)}\varepsilon_{14},\ y_1^{(q-1)}\varepsilon_{23},\ y_1^{(q-1)}\varepsilon_{34},\ y_1^{(q)}\varepsilon_\varnothing;$$
$$\text{rows } y_1^{(q-2)}\varepsilon_{134},\ y_1^{(q-1)}\varepsilon_1,\ y_1^{(q-1)}\varepsilon_2,\ y_1^{(q-1)}\varepsilon_3,\ y_1^{(q-1)}\varepsilon_4.$$

The only unit entries are those from $y_1^{(q)}\varepsilon_\varnothing$ to $y_1^{(q-1)}\varepsilon_1$ in each even degree $2q$, the edge $(y_1\varepsilon_\varnothing, \varepsilon_1)$ of the diagram repeated on every level. Canceling those pairs by a second matching leaves no unit entry, so

$$\beta_n^{\overline{R}}(R/I) = 1, 3, 4, 4, 4, \ldots \qquad \text{for } \overline{R} = R/(xy).$$

Example 8.2 (The running example, $c = 2$)**.** Add $a_2 = m_3 = zw$ and let $\sigma^{(2)}$ be Sobieska's homotopy for a_2, keeping $\sigma^{(1)} = \sigma$ of Example 8.1.

The nonzero values of $\sigma^{(2)}$ on cells are

$$\sigma^{(2)}(\varepsilon_\varnothing) = \varepsilon_3, \qquad \sigma^{(2)}(\varepsilon_1) = \varepsilon_{13}, \qquad \sigma^{(2)}(\varepsilon_2) = z\varepsilon_{23}, \qquad \sigma^{(2)}(\varepsilon_4) = -w\varepsilon_{34},$$
$$\sigma^{(2)}(\varepsilon_{12}) = z\varepsilon_{123}, \quad \sigma^{(2)}(\varepsilon_{14}) = -w\varepsilon_{134}, \quad \sigma^{(2)}(\varepsilon_{24}) = -zw\varepsilon_{234}, \quad \sigma^{(2)}(\varepsilon_{124}) = -zw\varepsilon_{1234}.$$

In matrices,

$$\sigma^{(2)}|_0 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad \sigma^{(2)}|_1 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & z & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -w \end{bmatrix},$$

$$\sigma^{(2)}|_2 = \begin{bmatrix} z & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -w & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -zw & 0 \end{bmatrix}, \quad \sigma^{(2)}|_3 = \begin{bmatrix} 0 & -zw & 0 & 0 \end{bmatrix}.$$

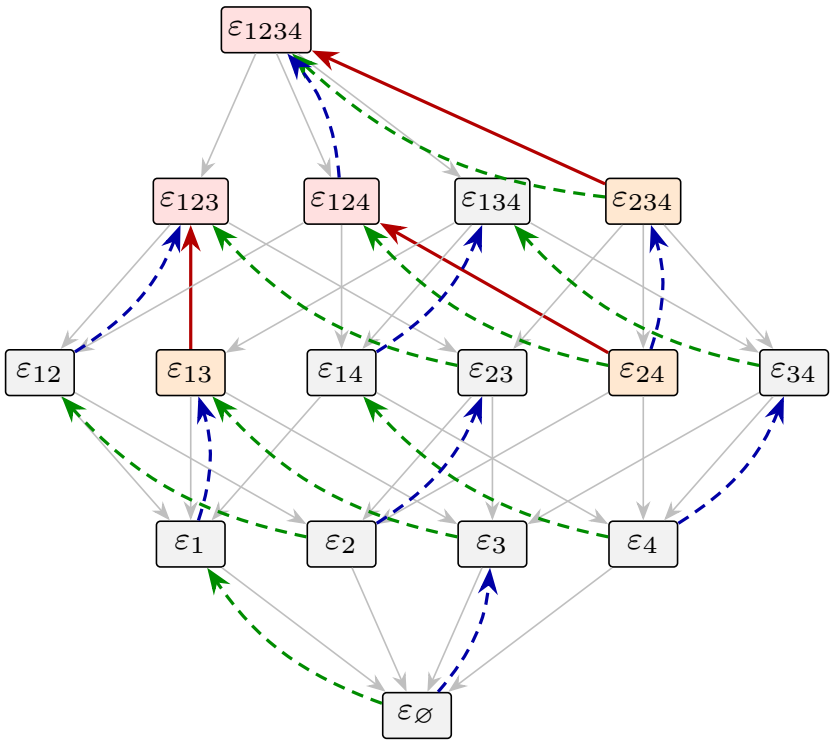


FIGURE 11. $G(F_\bullet)$ with the arrows of $\sigma^{(1)}$ and $\sigma^{(2)}$, $c = 2$.

We illustrate, the complex graph $G(F_\bullet)$ including the arrows induced by $\sigma^{(1)}$ and $\sigma^{(2)}$ in Figure 11.

Over $R/(xy, zw)$, $\mathrm{Sh}(F_\bullet, \sigma)$ has ranks $1, 4, 8, 12, 16, \ldots$, and d_{Sh} is block bidiagonal with connecting blocks $S_k = \sum_j T^{(k)}_{\boldsymbol{e}_j} \otimes \sigma^{(j)}$ (Proposition 4.1). With cells ordered by level and within a level by $\boldsymbol{u} = (k, 0), \ldots, (0, k)$, its matrices in degrees 2 and 3 are built from $S_1|_q = \begin{bmatrix} \sigma^{(1)}|_q & \sigma^{(2)}|_q \end{bmatrix}$:

$$d_{\mathrm{Sh}}|_2 = \left[\begin{array}{c|c} d_2 & S_1|_0 \end{array}\right] = \begin{array}{c} \begin{array}{cccccc|cc} \varepsilon_{12} & \varepsilon_{13} & \varepsilon_{14} & \varepsilon_{23} & \varepsilon_{24} & \varepsilon_{34} & y_1\varepsilon_\varnothing & y_2\varepsilon_\varnothing \end{array} \\ \left[\begin{array}{cccccc|cc} z & 0 & w & 0 & 0 & 0 & 1 & 0 \\ -x & 0 & 0 & w & xw & 0 & 0 & 0 \\ 0 & 0 & 0 & -y & 0 & x & 0 & 1 \\ 0 & 0 & -y & 0 & -yz & -z & 0 & 0 \end{array}\right] \end{array} \begin{array}{c} \\ \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \varepsilon_4 \end{array}$$

$$d_{\mathrm{Sh}}|_3 = \left[\begin{array}{c|c} d_3 & S_1|_1 \\ \hline 0 & \mathbf{1}_2 \otimes d_1 \end{array}\right]$$

which equals

$$\begin{array}{c} \begin{array}{cccc|cccc|cccc} \varepsilon_{123} & \varepsilon_{124} & \varepsilon_{134} & \varepsilon_{234} & y_1\varepsilon_1 & y_1\varepsilon_2 & y_1\varepsilon_3 & y_1\varepsilon_4 & y_2\varepsilon_1 & y_2\varepsilon_2 & y_2\varepsilon_3 & y_2\varepsilon_4 \end{array} \\ \left[\begin{array}{cccc|cccc|cccc} w & w & 0 & 0 & 0 & -y & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 & 0 & -1 & 0 & 1 & 0 & 0 & 0 \\ 0 & -z & -z & 0 & 0 & 0 & 0 & -x & 0 & 0 & 0 & 0 \\ x & 0 & 0 & x & 0 & 0 & 0 & 0 & 0 & z & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & y & y & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -w \\ \hline 0 & 0 & 0 & 0 & 0 & yz & 0 & xw & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & yz & 0 & xw \end{array}\right] \end{array} \begin{array}{c} \\ \varepsilon_{12} \\ \varepsilon_{13} \\ \varepsilon_{14} \\ \varepsilon_{23} \\ \varepsilon_{24} \\ \varepsilon_{34} \\ \hline y_1\varepsilon_\varnothing \\ y_2\varepsilon_\varnothing \end{array}$$

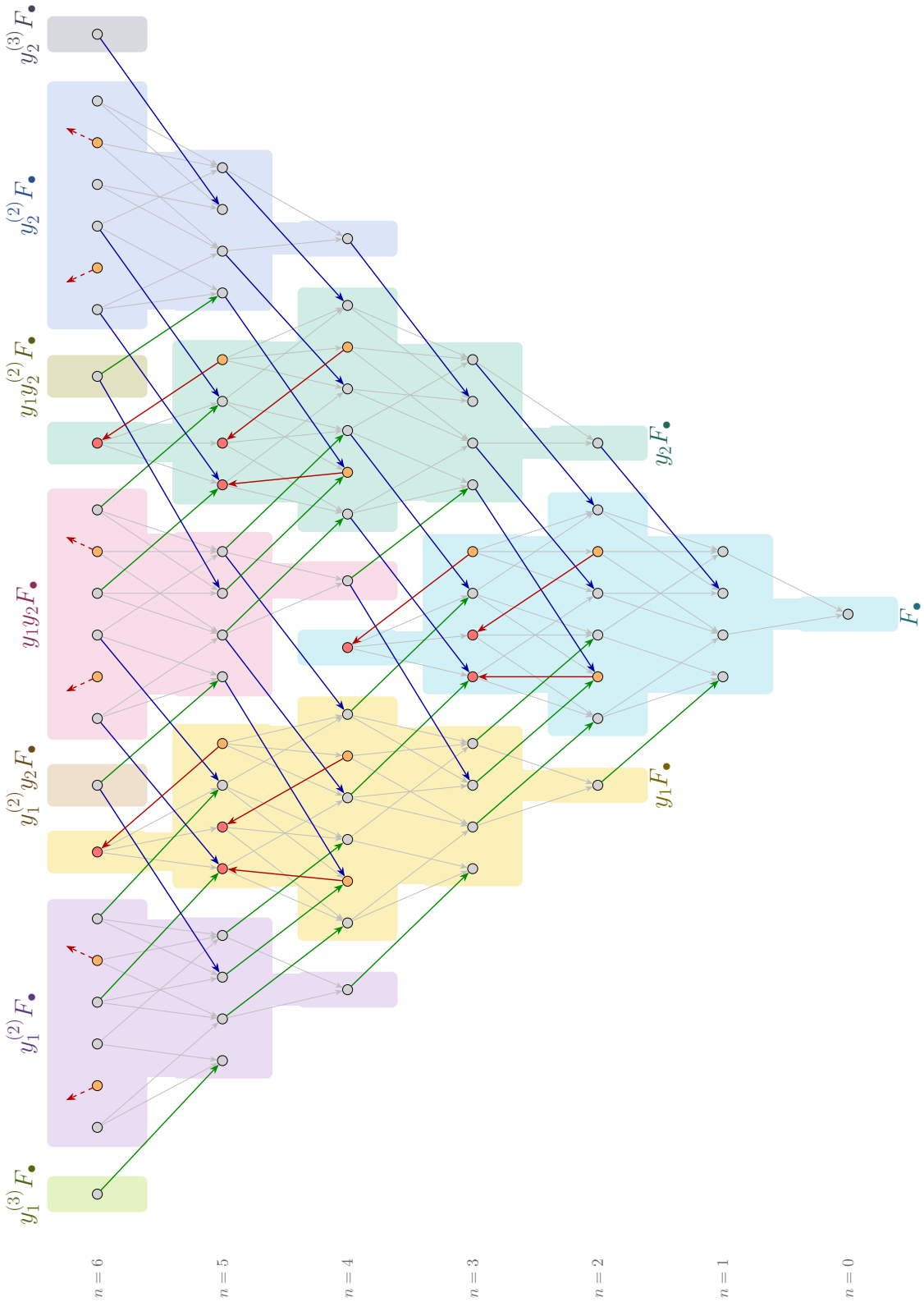


FIGURE 12. $G^{\mathrm{rep}}\big(\mathrm{Sh}(F_\bullet,\sigma),\mathscr{M}\big)$ through degree 6, $c=2$. Green: $\bar{\partial}_1$. Blue: $\bar{\partial}_2$.

Degree 4 is the first degree where an $S_2|_q = \left[\begin{array}{c|c|c} \sigma^{(1)}|_q & \sigma^{(2)}|_q & 0 \\ \hline 0 & \sigma^{(1)}|_q & \sigma^{(2)}|_q \end{array}\right]$ appears. We include below, but will not include the matrices in higher homological degree.

$$d_{\mathrm{Sh}}|_4 = \left[\begin{array}{c|c|c} d_4 & S_1|_2 & 0 \\ \hline 0 & \mathbf{1}_2 \otimes d_2 & S_2|_0 \end{array}\right]$$

which equals

ε_{1234}	$y_1\varepsilon_{12}$	$y_1\varepsilon_{13}$	$y_1\varepsilon_{14}$	$y_1\varepsilon_{23}$	$y_1\varepsilon_{24}$	$y_1\varepsilon_{34}$	$y_2\varepsilon_{12}$	$y_2\varepsilon_{13}$	$y_2\varepsilon_{14}$	$y_2\varepsilon_{23}$	$y_2\varepsilon_{24}$	$y_2\varepsilon_{34}$	$y_1^{(2)}\varepsilon_\varnothing$	$y_1y_2\varepsilon_\varnothing$	$y_2^{(2)}\varepsilon_\varnothing$	
1	0	0	0	y	0	0	z	0	0	0	0	0	0	0	0	ε_{123}
-1	0	0	0	0	xy	0	0	0	0	0	0	0	0	0	0	ε_{124}
1	0	0	0	0	0	x	0	0	$-w$	0	0	0	0	0	0	ε_{134}
-1	0	0	0	0	0	0	0	0	0	0	$-zw$	0	0	0	0	ε_{234}
0	z	0	w	0	0	0	0	0	0	0	0	0	1	0	0	$y_1\varepsilon_1$
0	$-x$	0	0	w	xw	0	0	0	0	0	0	0	0	0	0	$y_1\varepsilon_2$
0	0	0	0	$-y$	0	x	0	0	0	0	0	0	0	1	0	$y_1\varepsilon_3$
0	0	0	$-y$	0	$-yz$	$-z$	0	0	0	0	0	0	0	0	0	$y_1\varepsilon_4$
0	0	0	0	0	0	0	z	0	w	0	0	0	0	1	0	$y_2\varepsilon_1$
0	0	0	0	0	0	0	$-x$	0	0	w	xw	0	0	0	0	$y_2\varepsilon_2$
0	0	0	0	0	0	0	0	0	0	$-y$	0	x	0	0	1	$y_2\varepsilon_3$
0	0	0	0	0	0	0	0	0	$-y$	0	$-yz$	$-z$	0	0	0	$y_2\varepsilon_4$

As in Example 8.1, the edges of $\mathcal{M}$ are

$$\{(y_1^{(p)}y_2^{(q)}\varepsilon_{123}, y_1^{(p)}y_2^{(q)}\varepsilon_{13}), (y_1^{(p)}y_2^{(q)}\varepsilon_{124}, y_1^{(p)}y_2^{(q)}\varepsilon_{24}), (y_1^{(p)}y_2^{(q)}\varepsilon_{1234}, y_1^{(p)}y_2^{(q)}\varepsilon_{234})\}_{p,q\geq 0}.$$

There are also unit connecting edges $y_1^{(p)}y_2^{(q)}\varepsilon_\varnothing \xrightarrow{1} y_1^{(p-1)}y_2^{(q)}\varepsilon_1$ and $y_1^{(p)}y_2^{(q)}\varepsilon_3 \xrightarrow{-1} y_1^{(p-1)}y_2^{(q)}\varepsilon_{13}$ for $p \geq 1$, and $y_1^{(p)}y_2^{(q)}\varepsilon_\varnothing \xrightarrow{1} y_1^{(p)}y_2^{(q-1)}\varepsilon_3$ and $y_1^{(p)}y_2^{(q)}\varepsilon_1 \xrightarrow{1} y_1^{(p)}y_2^{(q-1)}\varepsilon_{13}$ for $q \geq 1$, shown in Figure 12.

As in Example 8.1, the transferred system is truncated. Its map $\sigma^{\mathcal{M}}_{\boldsymbol{e}_1}$ is the $\sigma^{\mathcal{M}}$ of Example 8.1, and $\sigma^{\mathcal{M}}_{\boldsymbol{e}_2}$ is given by

$$\sigma^{\mathcal{M}}_{\boldsymbol{e}_2}(\varepsilon_1) = p^{\mathcal{M}}(\varepsilon_{13}) = w\varepsilon_{12} + x\varepsilon_{23},$$

so

$$\sigma^{\mathcal{M}}_{\boldsymbol{e}_2}|_0 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \qquad \sigma^{\mathcal{M}}_{\boldsymbol{e}_2}|_1 = \begin{bmatrix} w & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ x & z & 0 & 0 \\ 0 & 0 & 0 & -w \end{bmatrix}, \qquad \sigma^{\mathcal{M}}_{\boldsymbol{e}_2}|_2 = \begin{bmatrix} 0 & -w & 0 & 0 \end{bmatrix}.$$

The reduced complex over $\overline{R} = R/(xy, zw)$ has ranks $1, 4, 6, 9, 11, 14, \ldots$. Its connecting block is $S_k = \sum_{j=1}^{2} T^{(k)}_{\boldsymbol{e}_j} \otimes \sigma^{\mathcal{M}}_{\boldsymbol{e}_j}$, the block of Proposition 4.1 for the base $F_\bullet{}^{\mathcal{M}}$, so that

$$S_1|_q = \left[\begin{array}{c|c} \sigma^{\mathcal{M}}_{\boldsymbol{e}_1}|_q & \sigma^{\mathcal{M}}_{\boldsymbol{e}_2}|_q \end{array}\right], \qquad S_2|_q = \left[\begin{array}{c|c|c} \sigma^{\mathcal{M}}_{\boldsymbol{e}_1}|_q & \sigma^{\mathcal{M}}_{\boldsymbol{e}_2}|_q & 0 \\ \hline 0 & \sigma^{\mathcal{M}}_{\boldsymbol{e}_1}|_q & \sigma^{\mathcal{M}}_{\boldsymbol{e}_2}|_q \end{array}\right],$$

and the matrices are $d^{\mathcal{M}}_{\mathrm{Sh}}|_1 = \left[d_1^{\mathcal{M}}\right] = \begin{matrix}\varepsilon_1 & \varepsilon_2 & \varepsilon_3 & \varepsilon_4\end{matrix}$
$\begin{bmatrix} 0 & yz & 0 & xw \end{bmatrix}$ $\varepsilon_\varnothing$,

$$d^{\mathcal{M}}_{\mathrm{Sh}}|_2 = \left[d_2^{\mathcal{M}} \mid S_1|_0\right] = \left[\begin{array}{cccc|cc} \varepsilon_{12} & \varepsilon_{14} & \varepsilon_{23} & \varepsilon_{34} & y_1\varepsilon_\varnothing & y_2\varepsilon_\varnothing \\ \hline z & w & 0 & 0 & 1 & 0 \\ -x & 0 & w & 0 & 0 & 0 \\ 0 & 0 & -y & x & 0 & 1 \\ 0 & -y & 0 & -z & 0 & 0 \end{array}\right] \begin{matrix}\varepsilon_1\\ \varepsilon_2\\ \varepsilon_3\\ \varepsilon_4,\end{matrix}$$

$$d^{\mathcal{M}}_{\mathrm{Sh}}|_3 = \left[\begin{array}{c|c} d_3^{\mathcal{M}} & S_1|_1 \\ \hline 0 & \mathbf{1}_2 \otimes d_1^{\mathcal{M}} \end{array}\right] = \left[\begin{array}{c|cccc|cccc} \varepsilon_{134} & y_1\varepsilon_1 & y_1\varepsilon_2 & y_1\varepsilon_3 & y_1\varepsilon_4 & y_2\varepsilon_1 & y_2\varepsilon_2 & y_2\varepsilon_3 & y_2\varepsilon_4 \\ \hline w & 0 & -y & -w & 0 & w & 0 & 0 & 0 \\ -z & 0 & 0 & 0 & -x & 0 & 0 & 0 & 0 \\ x & 0 & 0 & -x & 0 & x & z & 0 & 0 \\ y & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -w \\ \hline 0 & 0 & yz & 0 & xw & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & yz & 0 & xw \end{array}\right] \begin{matrix}\varepsilon_{12}\\ \varepsilon_{14}\\ \varepsilon_{23}\\ \varepsilon_{34}\\ \hline y_1\varepsilon_\varnothing\\ y_2\varepsilon_\varnothing,\end{matrix}$$

$$d^{\mathcal{M}}_{\mathrm{Sh}}|_4 = \left[\begin{array}{c|c} S_1|_2 & 0 \\ \hline \mathbf{1}_2 \otimes d_2^{\mathcal{M}} & S_2|_0 \end{array}\right]$$

which equals

$$\left[\begin{array}{cccc|cccc|c|c|c} y_1\varepsilon_{12} & y_1\varepsilon_{14} & y_1\varepsilon_{23} & y_1\varepsilon_{34} & y_2\varepsilon_{12} & y_2\varepsilon_{14} & y_2\varepsilon_{23} & y_2\varepsilon_{34} & y_1^{(2)}\varepsilon_\varnothing & y_1y_2\varepsilon_\varnothing & y_2^{(2)}\varepsilon_\varnothing & \\ \hline 0 & 0 & 0 & x & 0 & -w & 0 & 0 & 0 & 0 & 0 & \varepsilon_{134} \\ \hline z & w & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & y_1\varepsilon_1 \\ -x & 0 & w & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & y_1\varepsilon_2 \\ 0 & 0 & -y & x & 0 & 0 & 0 & 0 & 0 & 1 & 0 & y_1\varepsilon_3 \\ 0 & -y & 0 & -z & 0 & 0 & 0 & 0 & 0 & 0 & 0 & y_1\varepsilon_4 \\ \hline 0 & 0 & 0 & 0 & z & w & 0 & 0 & 0 & 1 & 0 & y_2\varepsilon_1 \\ 0 & 0 & 0 & 0 & -x & 0 & w & 0 & 0 & 0 & 0 & y_2\varepsilon_2 \\ 0 & 0 & 0 & 0 & 0 & 0 & -y & x & 0 & 0 & 1 & y_2\varepsilon_3 \\ 0 & 0 & 0 & 0 & 0 & -y & 0 & -z & 0 & 0 & 0 & y_2\varepsilon_4. \end{array}\right]$$

The unit entries are the 1 from $y^{\boldsymbol{u}}\varepsilon_\varnothing$ to $y^{\boldsymbol{u}-\boldsymbol{e}_1}\varepsilon_1$ whenever $u_1 \geq 1$ and to $y^{\boldsymbol{u}-\boldsymbol{e}_2}\varepsilon_3$ whenever $u_2 \geq 1$, the two unit edges $y_1\varepsilon_\varnothing \xrightarrow{1} \varepsilon_1$ and $y_2\varepsilon_\varnothing \xrightarrow{1} \varepsilon_3$ repeated over every $\boldsymbol{u}$. Matching each $y^{\boldsymbol{u}}\varepsilon_\varnothing$ with $|\boldsymbol{u}| \geq 1$ to one of them leaves no unit entry, so

$$\beta_n^{\overline{R}}(R/I) = 1, 2, 4, 6, 8, 10, \ldots, \quad \text{that is, } \beta_n^{\overline{R}}(R/I) = 2n \text{ for } n \geq 1, \qquad \text{for } \overline{R} = R/(xy, zw).$$

9. Algebraic Morse theory on the Shamash construction is again a Shamash construction

Throughout this section we fix a Morse matching $\mathcal{M}$ on $F_\bullet$, and split its differential as $d = \mu + \delta$. Let (i, p, h) and $(i^{\mathcal{M}}, p^{\mathcal{M}}, h^{\mathcal{M}})$ be the contractions of Theorem 6.8. Here i and p are the inclusion of and the projection onto the critical cells, and h is minus the inverse of μ on the lower cells. The contraction of $F_\bullet$ onto $F_\bullet{}^{\mathcal{M}}$ is then $(i^{\mathcal{M}}, p^{\mathcal{M}}, h^{\mathcal{M}}) = (\Phi i, p + p\delta\Phi h, \Phi h)$, where $\Phi = \sum_{n\geq 0}(h\delta)^n$.

We state the main theorem now to set up the hypotheses for the section, and postpone its proof until the end of the section.

Theorem 9.1. *Let $F_\bullet$ be a free resolution of R/I over R with a finite basis of each F_q, let $a_1, \ldots, a_c$ be an R-regular sequence in I, let $\sigma = \{\sigma^{(j)}\}_{j=1}^c$ be a truncated system of higher homotopies for $a_1, \ldots, a_c$ on $F_\bullet$, and let $\mathcal{M}$ be a Morse matching on $F_\bullet$.*

(i) *Under the identity map on the common basis* $\{y^{\boldsymbol{u}} \otimes \bar{\kappa}\}$, *with* $\boldsymbol{u} \in \mathbb{N}^c$ *and* κ *critical for* $\mathcal{M}$, $\mathrm{Sh}(F_\bullet, \sigma)^{\mathcal{M}}$ *is the free complex* $R\{y_j\}_{j=1}^c \otimes_R F_\bullet{}^{\mathcal{M}} \otimes_R \overline{R}$ *over* $\overline{R}$ *with differential*

$$d_{\mathrm{Sh}}^{\mathcal{M}} = \sum_{\boldsymbol{u} \in \mathbb{N}^c} t^{\boldsymbol{u}} \otimes \sigma_{\boldsymbol{u}}^{\mathcal{M}} \otimes \mathrm{id}\,.$$

(ii) *The maps* $\sigma^{\mathcal{M}}$ *form a system of higher homotopies for* $a_1, \dots, a_c$ *on* $F_\bullet{}^{\mathcal{M}}$ *over* R, *that is,* $\sigma_0^{\mathcal{M}} = d^{\mathcal{M}}$ *and*

$$\sum_{\boldsymbol{v}+\boldsymbol{v}'=\boldsymbol{u}} \sigma_{\boldsymbol{v}}^{\mathcal{M}} \sigma_{\boldsymbol{v}'}^{\mathcal{M}} = \begin{cases} a_j\,\mathrm{id} & \text{if } \boldsymbol{u} = \boldsymbol{e}_j, \\ 0 & \text{otherwise.} \end{cases}$$

So the complex in (i) is the Shamash resolution $\mathrm{Sh}(F_\bullet{}^{\mathcal{M}}, \sigma^{\mathcal{M}})$ *of Definition 2.4.*

The following corollary can be used to compute the ranks of the reduced Eisenbud-Shamash complex; in fact it was applied in Examples 8.1 and 8.2.

Corollary 9.2. *Let* $\mathcal{K}_q$ *be the set of cells of homological degree* q *that are critical for* $\mathcal{M}$, *the basis of the degree* q *part of* $K_\bullet$ *in Theorem 6.8, with* $\mathcal{K}_q = \emptyset$ *for* $q < 0$. *Then*

$$\operatorname{rank} \mathrm{Sh}(F_\bullet{}^{\mathcal{M}}, \sigma^{\mathcal{M}})_n = \sum_{\boldsymbol{u} \in \mathbb{N}^c} \#\mathcal{K}_{n-2|\boldsymbol{u}|},$$

and this is a free $\overline{R}$*-resolution of* R/I.

Proof. The rank count counts the basis $\{y^{\boldsymbol{u}} \otimes \bar{\kappa}\}$ of Theorem 9.1. For the resolution claim, $\mathrm{Sh}(F_\bullet, \sigma)$ is a free $\overline{R}$-resolution of R/I by Theorem 2.6, and $\mathrm{Sh}(F_\bullet{}^{\mathcal{M}}, \sigma^{\mathcal{M}})$ is its Morse complex along $\mathcal{M}$, which is homotopy equivalent to it. □

Remark 9.3. For monomial data as in Remark 5.6, the matched Eisenbud-Shamash complex is minimal exactly when no edge of its complex graph is a unit edge, in which case those ranks are the Betti numbers $\beta_n^{\overline{R}}(R/I)$.

For an R-linear map f between $F_\bullet$ and $F_\bullet{}^{\mathcal{M}}$ we write $\mathscr{f} = \mathrm{id} \otimes f$, so $\mathscr{f}(y^{\boldsymbol{u}} \otimes \alpha) = y^{\boldsymbol{u}} \otimes f(\alpha)$, with no sign, and $\mathscr{f}\mathscr{g} = \mathrm{id} \otimes fg$. These are maps over R, and the script letter marks the copy over each $\boldsymbol{u}$, as in the lifted matching $\mathcal{M}$. On $\mathrm{Sh}(F_\bullet, \sigma)$ and $\mathrm{Sh}(F_\bullet{}^{\mathcal{M}}, \sigma^{\mathcal{M}})$ we use their reductions $\overline{f}$. For the Greek maps δ and μ we write $\mathrm{id} \otimes \delta$ and $\mathrm{id} \otimes \mu$. We use $\mathscr{d}$, $\mathrm{id} \otimes \delta$, $\mathscr{d}^{\mathcal{M}}$, $(\mathscr{i}, \mathscr{p}, \mathscr{h})$ and $(\mathscr{i}^{\mathcal{M}}, \mathscr{p}^{\mathcal{M}}, \mathscr{h}^{\mathcal{M}})$, and $\mathcal{M}$ is the lifted matching.

Because the input is truncated, the transfer $\sigma_{\boldsymbol{u}}^{\mathcal{M}} = \sum \mathscr{p}^{\mathcal{M}} \sigma^{(j_1)} \mathscr{h}^{\mathcal{M}} \cdots \mathscr{h}^{\mathcal{M}} \sigma^{(j_k)} \mathscr{i}^{\mathcal{M}}$ of Definition 7.5 involves only the c maps $\sigma^{(j)}$, and in closed form it is the perturbation formula $d^{\mathcal{M}} = p\delta\Phi i$ of Theorem 6.8 with the connecting part ∂_{con} in place of δ.

Here $\mathscr{d} = \partial_0$ and $\partial_{\mathrm{con}} = \partial_1 + \cdots + \partial_c$ is the connecting part of Definition 3.1, so $D = \mathscr{d} + \partial_{\mathrm{con}}$, $d_{\mathrm{Sh}} = \overline{\mathscr{d}} + \bar{\partial}_{\mathrm{con}}$, and $\partial_{\mathrm{con}}(y^{\boldsymbol{u}} \otimes \alpha) = \sum_{j : u_j \geq 1} y^{\boldsymbol{u}-\boldsymbol{e}_j} \otimes \sigma^{(j)}(\alpha)$.

Every map in Proposition 9.4 is an R-linear map. The left side of the equality is the map D of the system of higher homotopies $\sigma^{\mathcal{M}}$, and its reduction modulo $\mathfrak{a}$ is the differential of $\mathrm{Sh}(F_\bullet{}^{\mathcal{M}}, \sigma^{\mathcal{M}})$. The identity is used over $\overline{R}$ only after reduction, in the proof of Theorem 9.1.

Proposition 9.4. *In the setting of Theorem 9.1, as operators from* $R\{y_j\}_{j=1}^c \otimes_R F_\bullet{}^{\mathcal{M}}$ *to itself,*

$$\sum_{\boldsymbol{u} \in \mathbb{N}^c} t^{\boldsymbol{u}} \otimes \sigma_{\boldsymbol{u}}^{\mathcal{M}} = \mathscr{d}^{\mathcal{M}} + \mathscr{p}^{\mathcal{M}} \partial_{\mathrm{con}} \Phi \mathscr{i}^{\mathcal{M}},$$

where $\Phi = \sum_{m \geq 0} (\mathscr{h}^{\mathcal{M}} \partial_{\mathrm{con}})^m$ *is a finite sum on each basis element.*

Proof. The term of the series with $m = k-1$ contributes

$$p^{\mathcal{M}}\partial_{\mathrm{con}}\big(\hbar^{\mathcal{M}}\partial_{\mathrm{con}}\big)^{k-1}i^{\mathcal{M}} = \sum_{(j_1,\dots,j_k)} \big(t_{j_1}\cdots t_{j_k}\big)\otimes\big(p^{\mathcal{M}}\sigma^{(j_1)}h^{\mathcal{M}}\cdots h^{\mathcal{M}}\sigma^{(j_k)}i^{\mathcal{M}}\big),$$

the sum being over all sequences of colors $(j_1,\dots,j_k)$ of length k. To observe this, substitute $\partial_{\mathrm{con}} = \sum_j t_j\otimes\sigma^{(j)}$ into each of the k factors and expand, using

$$(t\otimes f)(\mathrm{id}\otimes g) = t\otimes(fg),\ (\mathrm{id}\otimes g)(t\otimes f) = t\otimes(gf) \text{ and } (t\otimes f)(t'\otimes f') = (tt')\otimes(ff').$$

None of these carries a sign, because every t_j has even homological degree and id has homological degree 0.

Collecting the sequences with $\boldsymbol{e}_{j_1}+\cdots+\boldsymbol{e}_{j_k} = \boldsymbol{u}$ and using $t_{j_1}\cdots t_{j_k} = t^{\boldsymbol{u}}$ gives $\sum_{|\boldsymbol{u}|=k} t^{\boldsymbol{u}}\otimes\sigma^{\mathcal{M}}_{\boldsymbol{u}}$ by (2) of Definition 7.5. Summing over $k\geq 1$ and adding the term $\mathscr{d}^{\mathcal{M}} = \mathrm{id}\otimes\sigma^{\mathcal{M}}_0$ gives the display. The series Φ is finite on $y^{\boldsymbol{u}}\otimes\kappa$ because $\hbar^{\mathcal{M}}\partial_{\mathrm{con}}$ lowers $|\boldsymbol{u}|$ by one. □

So $\sigma^{\mathcal{M}}$ comes from ∂_{con} exactly as $d^{\mathcal{M}} = p\delta\Phi i$ comes from δ. Proposition 9.4 uses only the degrees of $i^{\mathcal{M}}$, $p^{\mathcal{M}}$, $h^{\mathcal{M}}$, so it holds for any such triple for which the series is finite, with $\sigma^{\mathcal{M}}_{\boldsymbol{u}}$ defined by Definition 7.5 (2) for that triple. For one color it is a single geometric series.

Corollary 9.5. *In the setting of Theorem 9.1 and Proposition 9.4, with $c = 1$, identify $\mathbb{N}^1$ with $\mathbb{N}$, and write $a = a_1$, $\sigma = \sigma^{(1)}$ and $\sigma^{\mathcal{M}}_k = \sigma^{\mathcal{M}}_{k\boldsymbol{e}_1}$. Then*

$$\sigma^{\mathcal{M}}_k = p^{\mathcal{M}}\sigma\big(h^{\mathcal{M}}\sigma\big)^{k-1}i^{\mathcal{M}}\quad(k\geq 1),\qquad \sum_{k\geq 1}\sigma^{\mathcal{M}}_k = p^{\mathcal{M}}\sigma\Phi i^{\mathcal{M}},\quad \Phi = \sum_{m\geq 0}(h^{\mathcal{M}}\sigma)^m.$$

So $\sigma^{\mathcal{M}}$ is truncated if and only if $p^{\mathcal{M}}\sigma(h^{\mathcal{M}}\sigma)^k i^{\mathcal{M}} = 0$ for every $k\geq 1$, and $\sigma h^{\mathcal{M}}\sigma = 0$ suffices.

The next lemma says the reduced resolution keeps the identity $D^2 = \Theta$, with $\Theta = \sum_j t_j\otimes a_j\,\mathrm{id}$, of Remark 3.2. Over $\overline{R}$ this would follow from the reduction alone, since Θ vanishes there, but over R the reduction must reproduce Θ. The idea is that the inclusion, perturbed along with the differential, still intertwines D and $D^{\mathcal{M}}$, and Θ, which commutes with all the maps involved, passes through it.

Lemma 9.6. *In the setting of Theorem 9.1, as operators from $R\{y_j\}_{j=1}^c\otimes_R F_\bullet^{\mathcal{M}}$ to itself, before the tensor with $\overline{R}$,*

$$\Big(\sum_{\boldsymbol{u}\in\mathbb{N}^c} t^{\boldsymbol{u}}\otimes\sigma^{\mathcal{M}}_{\boldsymbol{u}}\Big)\Big(\sum_{\boldsymbol{u}\in\mathbb{N}^c} t^{\boldsymbol{u}}\otimes\sigma^{\mathcal{M}}_{\boldsymbol{u}}\Big) = \sum_{j=1}^c t_j\otimes a_j\,\mathrm{id}\,.$$

Proof. Recall that $D = \mathscr{d} + \partial_{\mathrm{con}}$, so that $d_{\mathrm{Sh}} = \overline{D}$, and $\Theta = \sum_j t_j\otimes a_j\,\mathrm{id}$. By Proposition 9.4 the corresponding map for $\sigma^{\mathcal{M}}$, whose reduction is the differential in Theorem 9.1(i), is

$$D^{\mathcal{M}} = \sum_{\boldsymbol{u}\in\mathbb{N}^c} t^{\boldsymbol{u}}\otimes\sigma^{\mathcal{M}}_{\boldsymbol{u}} = \mathscr{d}^{\mathcal{M}} + p^{\mathcal{M}}\partial_{\mathrm{con}}\Phi i^{\mathcal{M}},\qquad \Phi = \sum_{m\geq 0}(\hbar^{\mathcal{M}}\partial_{\mathrm{con}})^m,$$

and we must show $(D^{\mathcal{M}})^2 = \Theta$. We use three facts.

(1) $D^2 = \Theta$ by Remark 3.2. Since $\mathscr{d}^2 = 0$, this says $\mathscr{d}\partial_{\mathrm{con}} + \partial_{\mathrm{con}}\mathscr{d} + \partial_{\mathrm{con}}\partial_{\mathrm{con}} = \Theta$.
(2) Θ commutes with ∂_{con} and with every map $f = \mathrm{id}\otimes f$, since the CI operators commute with each other and with every f, and each a_j lies in R.
(3) The contraction identities of $(i^{\mathcal{M}}, p^{\mathcal{M}}, h^{\mathcal{M}})$ hold for the maps $i^{\mathcal{M}}$, $p^{\mathcal{M}}$, $\hbar^{\mathcal{M}}$ between $R\{y_j\}_{j=1}^c\otimes_R F_\bullet$ and $R\{y_j\}_{j=1}^c\otimes_R F_\bullet^{\mathcal{M}}$ (these are R-linear maps, not attached to the lifted matching, which lives on $\mathrm{Sh}(F_\bullet,\sigma)$), since $f\mathscr{g} = \mathrm{id}\otimes fg$:

$$p^{\mathcal{M}}i^{\mathcal{M}} = \mathrm{id},\qquad \mathscr{d}\hbar^{\mathcal{M}} = i^{\mathcal{M}}p^{\mathcal{M}} - \mathrm{id} - \hbar^{\mathcal{M}}\mathscr{d},\qquad \hbar^{\mathcal{M}}\hbar^{\mathcal{M}} = \hbar^{\mathcal{M}}i^{\mathcal{M}} = p^{\mathcal{M}}\hbar^{\mathcal{M}} = 0,\qquad \mathscr{d}i^{\mathcal{M}} = i^{\mathcal{M}}\mathscr{d}^{\mathcal{M}}.$$

The series Φ is finite on each cell, since ∂_{con} lowers $|\boldsymbol{u}|$ by one and $\hbar^{\mathcal{M}}$ preserves it, and Φ inverts $\mathrm{id} - \hbar^{\mathcal{M}}\partial_{\mathrm{con}}$.

We perturb the inclusion along with the differential: set $\eta = \Phi i^{\mathcal{M}}$. Expanding the series Φ and factoring $\hbar^{\mathcal{M}}\partial_{\mathrm{con}}$ on the left from all terms with $m \geq 1$, we see that

$$\eta = \Phi i^{\mathcal{M}} = i^{\mathcal{M}} + \hbar^{\mathcal{M}}\partial_{\mathrm{con}}\Phi i^{\mathcal{M}} = i^{\mathcal{M}} + \hbar^{\mathcal{M}}\partial_{\mathrm{con}}\eta.$$

Replacing $\Phi i^{\mathcal{M}}$ with η in $D^{\mathcal{M}}$, we obtain $D^{\mathcal{M}} = \mathit{d}^{\mathcal{M}} + p^{\mathcal{M}}\partial_{\mathrm{con}}\eta$. We show

$$p^{\mathcal{M}}\eta = \mathrm{id} \qquad \text{and} \qquad D\eta = \eta D^{\mathcal{M}}.$$

Granting these, $\eta(D^{\mathcal{M}})^2 = D\eta D^{\mathcal{M}} = D^2\eta = \Theta\eta = \eta\Theta$ by (1) and (2), and applying $p^{\mathcal{M}}$ gives $(D^{\mathcal{M}})^2 = \Theta$.

$p^{\mathcal{M}}$ is a left inverse of η: By (3), $p^{\mathcal{M}}\eta = p^{\mathcal{M}}i^{\mathcal{M}} + p^{\mathcal{M}}\hbar^{\mathcal{M}}\partial_{\mathrm{con}}\eta = \mathrm{id}$.

η intertwines D and $D^{\mathcal{M}}$: Let $E = D\eta - \eta D^{\mathcal{M}}$. Replacing η by $i^{\mathcal{M}} + \hbar^{\mathcal{M}}\partial_{\mathrm{con}}\eta$ and using (3) for $\mathit{d}i^{\mathcal{M}}$ and $\mathit{d}\hbar^{\mathcal{M}}$,

$$\mathit{d}\eta = \mathit{d}i^{\mathcal{M}} + \mathit{d}\hbar^{\mathcal{M}}\partial_{\mathrm{con}}\eta = i^{\mathcal{M}}\mathit{d}^{\mathcal{M}} + i^{\mathcal{M}}p^{\mathcal{M}}\partial_{\mathrm{con}}\eta - \partial_{\mathrm{con}}\eta - \hbar^{\mathcal{M}}\mathit{d}\partial_{\mathrm{con}}\eta,$$

so $D\eta = \mathit{d}\eta + \partial_{\mathrm{con}}\eta = i^{\mathcal{M}}D^{\mathcal{M}} - \hbar^{\mathcal{M}}\mathit{d}\partial_{\mathrm{con}}\eta$.

Using the replacement $\eta = i^{\mathcal{M}} + \hbar^{\mathcal{M}}\partial_{\mathrm{con}}\eta$, we obtain $\eta D^{\mathcal{M}} = i^{\mathcal{M}}D^{\mathcal{M}} + \hbar^{\mathcal{M}}\partial_{\mathrm{con}}\eta D^{\mathcal{M}}$. Subtracting,

$$E = -\hbar^{\mathcal{M}}\big(\mathit{d}\partial_{\mathrm{con}}\eta + \partial_{\mathrm{con}}\eta D^{\mathcal{M}}\big) = -\hbar^{\mathcal{M}}\big(\mathit{d}\partial_{\mathrm{con}}\eta + \partial_{\mathrm{con}}D\eta\big) + \hbar^{\mathcal{M}}\partial_{\mathrm{con}}E,$$

using $\eta D^{\mathcal{M}} = D\eta - E$. By (1), $\mathit{d}\partial_{\mathrm{con}}\eta + \partial_{\mathrm{con}}D\eta = (\mathit{d}\partial_{\mathrm{con}} + \partial_{\mathrm{con}}\mathit{d} + \partial_{\mathrm{con}}\partial_{\mathrm{con}})\eta = \Theta\eta$, and by (2) and (3), $\hbar^{\mathcal{M}}\Theta\eta = \Theta\hbar^{\mathcal{M}}(i^{\mathcal{M}} + \hbar^{\mathcal{M}}\partial_{\mathrm{con}}\eta) = 0$. So $(\mathrm{id} - \hbar^{\mathcal{M}}\partial_{\mathrm{con}})E = 0$, and applying Φ gives $E = 0$. □

The following lemma is a version of [25, Proposition 3.6] under weaker hypotheses. Real assumes that each of δ_1, δ_2 and $\delta_1 + \delta_2$ is a perturbation of $(N_\bullet, d)$ whose composite with h is pointwise nilpotent, and concludes that the two perturbed contractions agree. We do not need δ_2 to be a perturbation of $(N_\bullet, d)$ on its own, which need not hold in our application, and we only compare the perturbed differentials.

Lemma 9.7 (Perturbation Additivity Lemma)**.** *Let (i, p, h) be a contraction of a complex $(N_\bullet, d)$ onto a complex $(N'_\bullet, d')$ of A-modules, and let $\delta_1, \delta_2 \colon N_\bullet \to N_{\bullet-1}$ be A-linear maps such that δ_1 and $\delta_1 + \delta_2$ are perturbations of $(N_\bullet, d)$. Suppose that*

$$\Phi_1 = \sum_{n\geq 0}(h\delta_1)^n = (\mathrm{id} - h\delta_1)^{-1} \text{ and } \Phi_{12} = \sum_{n\geq 0}\big(h(\delta_1+\delta_2)\big)^n = \big(\mathrm{id} - h(\delta_1+\delta_2)\big)^{-1}$$

are finite sums on each element, let

$$(i_1, p_1, h_1) = \big(\Phi_1 i,\ p + p\delta_1\Phi_1 h,\ \Phi_1 h\big)$$

be the contraction of $(N_\bullet, d + \delta_1)$ onto $(N'_\bullet, d' + p\delta_1\Phi_1 i)$ of Lemma 6.7, and suppose that

$$\Phi_2 = \sum_{n\geq 0}(h_1\delta_2)^n = (\mathrm{id} - h_1\delta_2)^{-1}$$

is a finite sum on each element. Then perturbing by δ_1 and then by δ_2 gives the same differential on $N'_\bullet$ as perturbing by $\delta_1 + \delta_2$ at once:

$$d' + p(\delta_1+\delta_2)\Phi_{12}i = d' + p\delta_1\Phi_1 i + p_1\delta_2\Phi_2 i_1.$$

Proof. Figure 13 shows the two ways of perturbing by $\delta_1 + \delta_2$. In Figure 13 (i_2, p_2, h_2) is the contraction of Lemma 6.7 for (i_1, p_1, h_1) and δ_2, which is a perturbation of $(N_\bullet, d + \delta_1)$ since $(d + \delta_1 + \delta_2)^2 = 0$, and (i_{12}, p_{12}, h_{12}) the one for (i, p, h) and $\delta_1 + \delta_2$. So every row is a contraction, and it remains to compare the two perturbation terms.

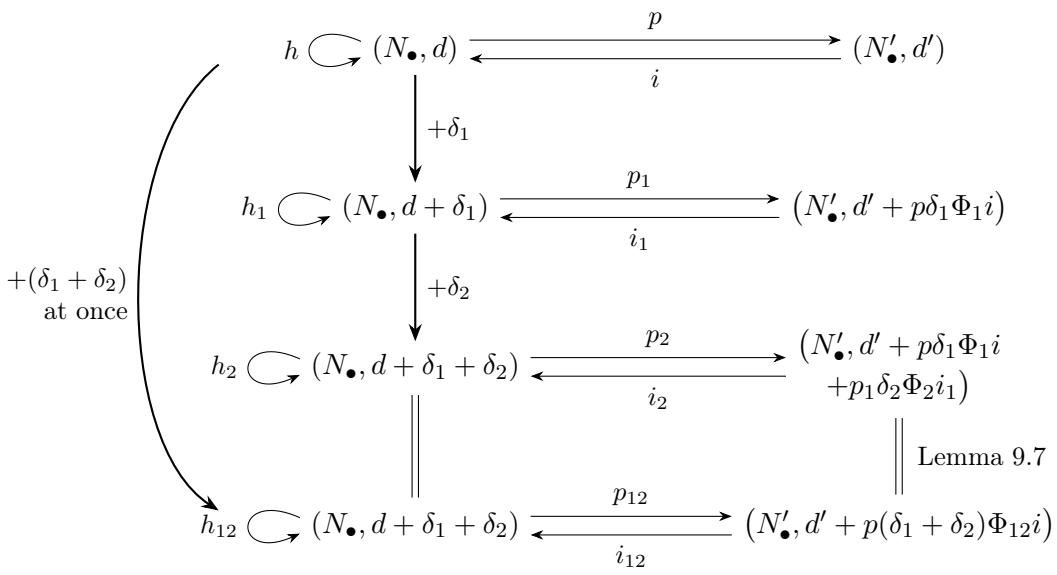


FIGURE 13. Perturbing by δ_1 and then by δ_2, or by $\delta_1 + \delta_2$ at once.

Since $(\mathrm{id} - h\delta_1)\Phi_1 = \mathrm{id}$ and $h_1 = \Phi_1 h$,

$$\mathrm{id} - h(\delta_1+\delta_2) = (\mathrm{id} - h\delta_1)(\mathrm{id} - \Phi_1 h\delta_2) = (\mathrm{id} - h\delta_1)(\mathrm{id} - h_1\delta_2),$$

so $\Phi_{12} = \Phi_2\Phi_1$. The left side and right side respectively are

$$p\delta_1\Phi_2\Phi_1 i + p\delta_2\Phi_2\Phi_1 i \text{ and } p\delta_1\Phi_1 i + p\delta_2\Phi_2\Phi_1 i + p\delta_1\Phi_1 h\delta_2\Phi_2\Phi_1 i.$$

Their difference is

$$p\delta_1(\Phi_2 - \mathrm{id})\Phi_1 i - p\delta_1\Phi_1 h\delta_2\Phi_2\Phi_1 i,$$

which is zero because $\Phi_2 - \mathrm{id} = h_1\delta_2\Phi_2 = \Phi_1 h\delta_2\Phi_2$. □

The Perturbation Additivity Lemma is the main tool for the proof of the main theorem, and we are now ready to begin.

Proof of Theorem 9.1. Part (i): The edges of $\mathcal{M}$ lie over those of $\mathcal{M}$, so the matched part of d_{Sh} is $\mathrm{id}\otimes\bar{\mu}$ and the unmatched part is $\mathrm{id}\otimes\bar{\delta} + \bar{\partial}_{\mathrm{con}}$. Theorem 6.8 attaches to $\mathcal{M}$ the contraction $(\bar{i}, \bar{p}, \bar{h})$, and since $\mathcal{M}$ is a Morse matching (Proposition 7.3), the theorem applies and gives

$$d_{\mathrm{Sh}}^{\mathcal{M}} = \bar{p}\,(\mathrm{id}\otimes\bar{\delta} + \bar{\partial}_{\mathrm{con}})\,\overline{\Phi}_{12}\,\bar{i}, \qquad \Phi_{12} = \sum_{n\geq 0}\big(\hbar(\mathrm{id}\otimes\delta + \partial_{\mathrm{con}})\big)^n,$$

where we use $\bar{f}\,\bar{g} = \overline{fg}$ for R-linear maps f and g.

In other words, $\mathrm{Sh}(F_\bullet, \sigma)^{\mathcal{M}}$ comes from perturbing $\mathrm{id}\otimes\bar{\mu}$ by $\mathrm{id}\otimes\bar{\delta} + \bar{\partial}_{\mathrm{con}}$ all at once. We show that perturbing first by $\mathrm{id}\otimes\bar{\delta}$ and then by $\bar{\partial}_{\mathrm{con}}$ gives the same result, as in Figure 14. This is exactly the situation of Lemma 9.7: Figure 14 is Figure 13 for the contraction $(\bar{i}, \bar{p}, \bar{h})$, with $d = \mathrm{id}\otimes\bar{\mu}$, $d' = 0$, $\delta_1 = \mathrm{id}\otimes\bar{\delta}$ and $\delta_2 = \bar{\partial}_{\mathrm{con}}$. In particular (i_2, p_2, h_2) is the contraction obtained by perturbing by δ_1 and then by δ_2.

The maps in the top two rows of Figure 14 are of the form $\overline{f}$ for maps f on $F_\bullet$, since $\mathcal{M}$ and $\mathrm{id}\otimes\bar{\delta}$ are copies of $\mathcal{M}$ and $\bar{\delta}$ over each $\boldsymbol{u}$. This fails in the bottom two rows, because ∂_{con} lowers $|\boldsymbol{u}|$ and is not of the form f. The proof of Lemma 9.7 also gives $(i_2, p_2, h_2) = (i^{\mathcal{M}}, p^{\mathcal{M}}, h^{\mathcal{M}})$, although we do not need this.

Lemma 9.7 says that perturbing by δ_1 and then by δ_2 gives the same differential,

$$p\delta_1\Phi_1 i + p_1\delta_2\Phi_2 i_1,$$

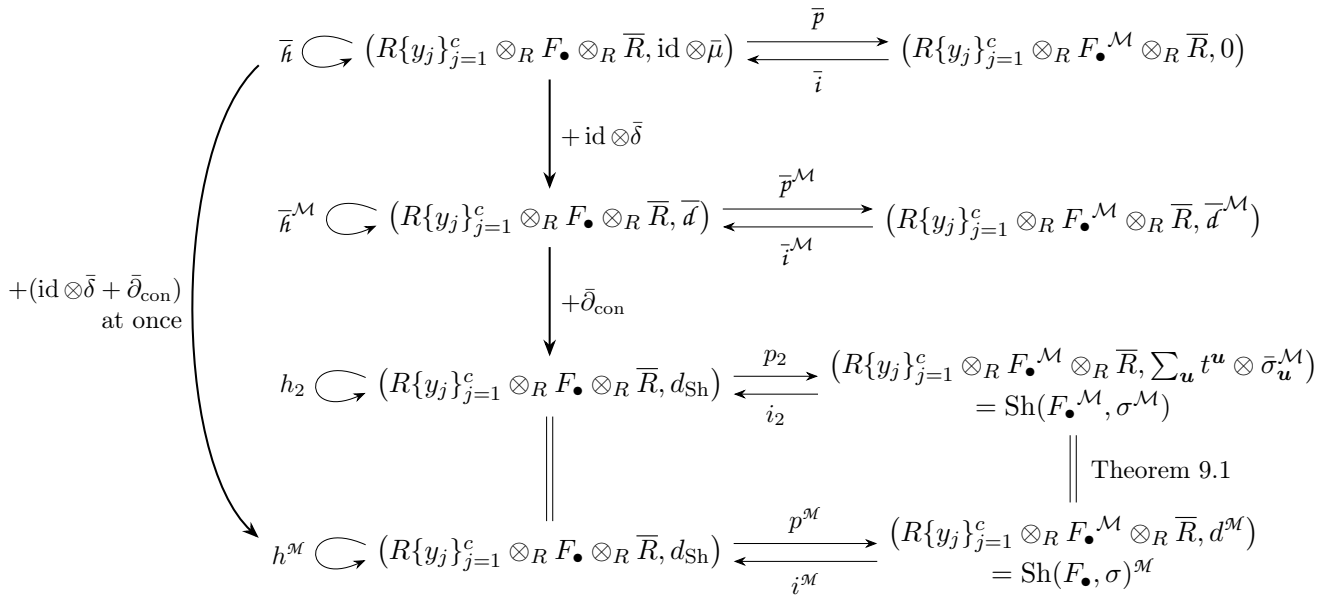


FIGURE 14. Matching done in two ways.

as perturbing by $\delta_1+\delta_2$ at once. We apply it over $\overline{R}$ to the contraction $(\bar{i},\overline{p},\overline{h})$ of

$$\big(R\{y_j\}_{j=1}^c \otimes_R F_\bullet \otimes_R \overline{R}, \mathrm{id}\otimes\bar{\mu}\big) \text{ onto } \big(R\{y_j\}_{j=1}^c \otimes_R K_\bullet \otimes_R \overline{R}, 0\big),$$

with $\delta_1 = \mathrm{id}\otimes\bar{\delta}$ and $\delta_2 = \bar{\partial}_{\mathrm{con}}$. Both δ_1 and $\delta_1+\delta_2$ are perturbations, since

$$(\mathrm{id}\otimes\bar{\mu}+\mathrm{id}\otimes\bar{\delta})^2 = \overline{d}^{\,2} = 0 \quad\text{and}\quad (\mathrm{id}\otimes\bar{\mu}+\mathrm{id}\otimes\bar{\delta}+\bar{\partial}_{\mathrm{con}})^2 = \overline{D^2} = d_{\mathrm{Sh}}^2 = 0,$$

the second because every a_j is zero in $\overline{R}$. The series in the lemma are $\overline{\Phi}_{12}$, $\overline{\Phi}_1$ and $\overline{\Phi}_2$, with Φ_{12} as above, $\Phi_1 = \sum_{n\ge0}\big(\hbar(\mathrm{id}\otimes\delta)\big)^n$ and $\Phi_2 = \sum_{n\ge0}(\hbar^{\mathcal{M}}\partial_{\mathrm{con}})^n$. Each is finite on each cell: Φ_{12} as above, Φ_1 by Theorem 6.8 for $\mathcal{M}$, and Φ_2 by Proposition 9.4, since ∂_{con} lowers $|\boldsymbol{u}|$. Perturbing by δ_1 alone is just Theorem 6.8 for $\mathcal{M}$, applied in each copy $y^{\boldsymbol{u}}\otimes F_\bullet$. That theorem gives $d^{\mathcal{M}} = p\delta\Phi i$ and $(i^{\mathcal{M}}, p^{\mathcal{M}}, h^{\mathcal{M}}) = (\Phi i, p+p\delta\Phi h, \Phi h)$ with $\Phi = \sum_{n\ge0}(h\delta)^n$, so lifting and then reducing gives

$$p\,(\mathrm{id}\otimes\delta)\,\Phi_1\,i = \overline{d}^{\mathcal{M}}, \qquad (i_1,p_1,h_1) = (\bar{i}^{\mathcal{M}}, \overline{p}^{\mathcal{M}}, \overline{h}^{\mathcal{M}}),$$

Now Lemma 9.7 and Proposition 9.4, tensored with $\overline{R}$, give

$$d^{\mathcal{M}}_{\mathrm{Sh}} = \overline{d}^{\mathcal{M}} + \overline{p}^{\mathcal{M}}\,\bar{\partial}_{\mathrm{con}}\,\overline{\Phi}_2\,\bar{i}^{\mathcal{M}} = \sum_{\boldsymbol{u}\in\mathbb{N}^c} t^{\boldsymbol{u}}\otimes\bar{\sigma}^{\mathcal{M}}_{\boldsymbol{u}},$$

which is the Shamash differential of the system $\sigma^{\mathcal{M}}$ on the cells $y^{\boldsymbol{u}}\otimes\bar{\kappa}$ critical for $\mathcal{M}$.

Part (ii): Expanding the square in Lemma 9.6 gives

$$\sum_{\boldsymbol{u}} t^{\boldsymbol{u}}\otimes\Big(\sum_{\boldsymbol{u_1}+\boldsymbol{u_2}=\boldsymbol{u}}\sigma^{\mathcal{M}}_{\boldsymbol{u_1}}\sigma^{\mathcal{M}}_{\boldsymbol{u_2}}\Big) = \sum_j t_j\otimes a_j\,\mathrm{id}\,.$$

On $y^{\boldsymbol{v}}\otimes\kappa$ the terms for distinct $\boldsymbol{u}\le\boldsymbol{v}$ lie in distinct summands $y^{\boldsymbol{v}-\boldsymbol{u}}\otimes F_\bullet{}^{\mathcal{M}}$, so we can compare the coefficients of each $t^{\boldsymbol{u}}$, and

$$\sum_{\boldsymbol{u_1}+\boldsymbol{u_2}=\boldsymbol{u}}\sigma^{\mathcal{M}}_{\boldsymbol{u_1}}\sigma^{\mathcal{M}}_{\boldsymbol{u_2}} = \begin{cases} a_j\,\mathrm{id} & \text{when } \boldsymbol{u}=\boldsymbol{e}_j\\ 0 & \text{otherwise.}\end{cases}$$

□

Remark 9.8. By Theorem 9.1, $\mathrm{Sh}(F_\bullet{}^{\mathcal{M}}, \sigma^{\mathcal{M}})$ is a free $\overline{R}$-resolution of R/I. Its level edges have the weights of the edges of $G(F_\bullet{}^{\mathcal{M}})$, and its color-$j$ connecting edges have the weights of the entries of $\overline{(\sigma^{\mathcal{M}})^{(j)}}$. So when a connecting edge has unit weight, $\mathrm{Sh}(F_\bullet{}^{\mathcal{M}}, \sigma^{\mathcal{M}})$ is not minimal, even when $F_\bullet{}^{\mathcal{M}}$ is, and reducing along a lifted matching does not produce the minimal free resolution of R/I over $\overline{R}$. Such edges can still be reduced by Proposition 7.1.

Remark 9.9. The transferred maps need not square to zero: by Theorem 9.1(ii),

$$\sigma^{\mathcal{M}}_{\boldsymbol{e}_j}\sigma^{\mathcal{M}}_{\boldsymbol{e}_j} = p^{\mathcal{M}}\sigma^{(j)}(dh^{\mathcal{M}} + h^{\mathcal{M}}d)\sigma^{(j)}i^{\mathcal{M}} = -(d^{\mathcal{M}}\sigma^{\mathcal{M}}_{2\boldsymbol{e}_j} + \sigma^{\mathcal{M}}_{2\boldsymbol{e}_j}d^{\mathcal{M}}).$$

This is the same thing we saw for the $c = 2$ lattice of Figure 2, where a general system has $\sigma^{(1)}\sigma^{(2)} + \sigma^{(2)}\sigma^{(1)} = -(d\sigma_{(1,1)} + \sigma_{(1,1)}d)$ and the diagonal arrow of $\sigma_{(1,1)}$ accounts for the difference. Here the arrow of $\sigma^{\mathcal{M}}_{2\boldsymbol{e}_j}$, which jumps two levels, plays that role. The transferred system is truncated if

$$\sigma^{(j)}h^{\mathcal{M}}\sigma^{(k)} = 0 \qquad \text{for all colors } j, k, \tag{3}$$

since every summand of $\sigma^{\mathcal{M}}_{\boldsymbol{u}}$ with $|\boldsymbol{u}| \geq 2$ contains a factor $\sigma^{(j)}h^{\mathcal{M}}\sigma^{(k)}$.

If no entry of δ joins an upper cell to a lower cell, then $h\delta h = 0$, so $h^{\mathcal{M}} = h$ and

$$(\sigma^{(j)}h\sigma^{(k)})(\gamma) = -\sum_{(\alpha,\beta)\in\mathcal{M}} d^{-1}_{\alpha,\beta}\sigma^{(k)}_{\gamma,\beta}\sigma^{(j)}(\alpha),$$

with α upper and β lower. So (3) holds as soon as every pair $(\alpha, \beta) \in \mathcal{M}$ satisfies one of the following.

(1) No $\sigma^{(k)}$ has a nonzero entry into β.

(2) Every $\sigma^{(j)}$ vanishes on α.

That is, no gradient path in $\mathrm{Sh}(F_\bullet, \sigma)$ leaves a cell along a connecting edge, crosses one reversed matched edge, and leaves along another connecting edge. Without $h^{\mathcal{M}} = h$ the path may cross several reversed matched edges, as in the next example.

Example 9.10. This example shows that the transferred system need not be truncated. Let $I = (x^3, x^2y, xy^2, y^3) \subset \Bbbk[x, y]$, and let $F_\bullet$ be its Taylor resolution on the generators $m_1, \ldots, m_4$ in that order. We take $c = 1$ and $a = x^3 = m_1$. Sobieska's homotopy (Section 5) for the one term expression $a = m_1$ is the truncated system

$$\sigma(\varepsilon_S) = (-1)^{\#\{s\in S : s>1\}}\gcd(m_1, m_S)\varepsilon_{S\cup 1} \quad (1 \notin S), \qquad \sigma(\varepsilon_S) = 0 \quad (1 \in S).$$

We match ε_{124} with ε_{14} and ε_{234} with ε_{24}, two edges of weight -1. Since $G^{\mathrm{rep}}(F_\bullet, \mathcal{M})$ has no directed cycle, $\mathcal{M}$ is a Morse matching. We leave $(\varepsilon_{123}, \varepsilon_{13})$ and $(\varepsilon_{1234}, \varepsilon_{134})$ unmatched on purpose. Matching them as well would reach the minimal resolution, which has length two, and there the transferred system is truncated by Remark 2.3.

The critical cells are $\varepsilon_\varnothing$, then $\varepsilon_1, \varepsilon_2, \varepsilon_3, \varepsilon_4$, then $\varepsilon_{12}, \varepsilon_{13}, \varepsilon_{23}, \varepsilon_{34}$, then $\varepsilon_{123}, \varepsilon_{134}$, and ε_{1234}, so $F_\bullet{}^{\mathcal{M}}$ has ranks $1, 4, 4, 2, 1$. In these bases its differential is

$$d^{\mathcal{M}}_1 = \begin{bmatrix} x^3 & x^2y & xy^2 & y^3 \end{bmatrix}, \qquad d^{\mathcal{M}}_2 = \begin{bmatrix} y & y^2 & 0 & 0 \\ -x & 0 & y & 0 \\ 0 & -x^2 & -x & y \\ 0 & 0 & 0 & -x \end{bmatrix},$$

$$d^{\mathcal{M}}_3 = \begin{bmatrix} y & -y^2 \\ -1 & y \\ x & -xy \\ 0 & 0 \end{bmatrix}, \qquad d^{\mathcal{M}}_4 = \begin{bmatrix} y \\ 1 \end{bmatrix}.$$

and the transferred system is

$$\sigma_1^{\mathcal{M}}|_0 = \begin{bmatrix}1\\0\\0\\0\end{bmatrix}, \quad \sigma_1^{\mathcal{M}}|_1 = \begin{bmatrix}0 & -x^2 & 0 & -y^2\\0 & 0 & -x & 0\\0 & 0 & 0 & -xy\\0 & 0 & 0 & -x^2\end{bmatrix}, \quad \sigma_1^{\mathcal{M}}|_2 = \begin{bmatrix}0 & 0 & x^2 & 0\\0 & 0 & 0 & x\end{bmatrix}, \quad \sigma_1^{\mathcal{M}}|_3 = \begin{bmatrix}0 & x^3\end{bmatrix},$$

$$\sigma_2^{\mathcal{M}}|_1 = \begin{bmatrix}0 & 0 & 0 & x^3\end{bmatrix} : F_1^{\mathcal{M}} \to F_4^{\mathcal{M}}, \qquad \sigma_2^{\mathcal{M}}|_q = 0 \quad (q \neq 1).$$

Only one path contributes to $\sigma_2^{\mathcal{M}} = p^{\mathcal{M}}\sigma h^{\mathcal{M}}\sigma i^{\mathcal{M}}$. Starting from ε_4, the map σ goes up to the lower cell $-\varepsilon_{14}$, then $h^{\mathcal{M}}$ goes up to $-\varepsilon_{124} - x\varepsilon_{234}$, and σ goes up again, so

$$\sigma_2^{\mathcal{M}}(\varepsilon_4) = \sigma\big(-\varepsilon_{124} - x\varepsilon_{234}\big) = x^3\varepsilon_{1234}.$$

This path crosses two reversed matched edges between its two connecting edges. Each matched pair satisfies one of the conditions of Remark 9.9, (2) at $(\varepsilon_{124}, \varepsilon_{14})$ and (1) at $(\varepsilon_{234}, \varepsilon_{24})$. Those conditions need $h^{\mathcal{M}} = h$, however, and here the entry x of δ from ε_{124} to ε_{24} makes $h^{\mathcal{M}} \neq h$.

Finally, Theorem 9.1(ii) gives $(\sigma_1^{\mathcal{M}})^2 = -(d^{\mathcal{M}}\sigma_2^{\mathcal{M}} + \sigma_2^{\mathcal{M}}d^{\mathcal{M}})$, and here both sides are nonzero. They vanish except on $F_1^{\mathcal{M}}$ and $F_2^{\mathcal{M}}$, where

$$(\sigma_1^{\mathcal{M}})^2|_1 = \begin{bmatrix}0 & 0 & 0 & -x^3y\\0 & 0 & 0 & -x^3\end{bmatrix}, \qquad \big(d^{\mathcal{M}}\sigma_2^{\mathcal{M}} + \sigma_2^{\mathcal{M}}d^{\mathcal{M}}\big)\big|_1 = \begin{bmatrix}0 & 0 & 0 & x^3y\\0 & 0 & 0 & x^3\end{bmatrix},$$

$$(\sigma_1^{\mathcal{M}})^2|_2 = \begin{bmatrix}0 & 0 & 0 & x^4\end{bmatrix}, \qquad \big(d^{\mathcal{M}}\sigma_2^{\mathcal{M}} + \sigma_2^{\mathcal{M}}d^{\mathcal{M}}\big)\big|_2 = \begin{bmatrix}0 & 0 & 0 & -x^4\end{bmatrix}.$$

Over $\overline{R} = R/(x^3)$ all of these entries are zero, so the failure of truncation can only be seen over R.

In Examples 8.1 and 8.2 the transferred system is truncated, while in Example 9.10 it is not, even though there $\operatorname{pd}_R R/I = 2$. So whether $\sigma^{\mathcal{M}}$ is truncated depends on the matching, and not only on the ideal.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Claude (Anthropic) to make the diagrams and to help spot mathematical errors throughout the paper. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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